

DSC 12G-HD A and DSC HD-12G A

12G-SDI and HDMI Scalers with Stereo Audio



User Guide Video Scalers

Safety Instructions

Safety Instructions • English

⚠ WARNING: This symbol, , when used on the product, is intended to alert the user of the presence of uninsulated dangerous voltage within the product's enclosure that may present a risk of electric shock.

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Para obtener información sobre directrices de seguridad, cumplimiento de normativas, compatibilidad electromagnética, accesibilidad y temas relacionados, consulte la Guía de cumplimiento de normativas y seguridad de Extron, referencia 68-290-01, en el sitio Web de Extron, www.extron.com.

Instructions de sécurité • Français

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⚠ ATTENTION : Ce pictogramme, , lorsqu'il est utilisé sur le produit, signale à l'utilisateur des instructions d'utilisation ou de maintenance importantes qui se trouvent dans la documentation fournie avec l'équipement.

Pour en savoir plus sur les règles de sécurité, la conformité à la réglementation, la compatibilité EMI/EMF, l'accessibilité, et autres sujets connexes, lisez les informations de sécurité et de conformité Extron, réf. 68-290-01, sur le site Extron, ■ www.extron.com.

Istruzioni di sicurezza • Italiano

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⚠ ATTENZIONE: Il simbolo, , se usato sul prodotto, serve ad avvertire l'utente della presenza di importanti istruzioni di funzionamento e manutenzione nella documentazione fornita con l'apparecchio.

Per informazioni su parametri di sicurezza, conformità alle normative, compatibilità EMI/EMF, accessibilità e argomenti simili, fare riferimento alla Guida alla conformità normativa e di sicurezza di Extron, cod. articolo 68-290-01, sul sito web di Extron, www.extron.com.

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Для получения информации о правилах техники безопасности, соблюдении нормативных требований, электромагнитной совместимости (ЭМП/ЭДС), возможности доступа и других вопросах см. руководство по безопасности и соблюдению нормативных требований Extron на сайте Extron: www.extron.com, номер по каталогу - 68-290-01.

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有關安全性指導方針、法規遵守、EMI/EMF 相容性、存取範圍和相關主題的詳細資訊，請瀏覽 Extron 網站：www.extron.com，然後參閱《Extron 安全性與法規遵守手冊》，準則編號 68-290-01。

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安全上のご注意、法規遵守、EMI/EMF適合性、その他の関連項目については、エクストロンのウェブサイト www.extron.com より「Extron Safety and Regulatory Compliance Guide」(P/N 68-290-01) をご覧ください。

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경고: 이 기호 ⚠가 제품에 사용될 경우, 제품의 인클로저 내에 있는 접지되지 않은 위험한 전류로 인해 사용자가 감전될 위험이 있음을 경고합니다.

주의: 이 기호 ⚠가 제품에 사용될 경우, 장비와 함께 제공된 책자에 나와 있는 주요 운영 및 유지보수(정비) 지침을 경고합니다.

안전 가이드라인, 규제 준수, EMI/EMF 호환성, 접근성, 그리고 관련 항목에 대한 자세한 내용은 Extron 웹 사이트(www.extron.com)의 Extron 안전 및 규제 준수 안내서, 68-290-01 조항을 참조하십시오.

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This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC rules. The Class A limits provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause interference. This interference must be corrected at the expense of the user.

NOTE: For more information on safety guidelines, regulatory compliances, EMI/EMF compatibility, accessibility, and related topics, see the [Extron Safety and Regulatory Compliance Guide](#) on the Extron website.

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VCCI-A

Battery Notice

This product contains a battery. **Do not open the unit to replace the battery.** If the battery needs replacing, return the entire unit to Extron (for the correct address, see the Extron Warranty section on the last page of this guide).

CAUTION: Risk of explosion. Do not replace the battery with an incorrect type. Dispose of used batteries according to the instructions.

ATTENTION : Risque d'explosion. Ne pas remplacer la pile par le mauvais type de pile. Débarrassez-vous des piles usagées selon le mode d'emploi.

Remarque : Pour plus d'informations sur les directives de sécurité, les conformités de régulation, la compatibilité EMI/EMF, l'accessibilité, et les sujets en lien, consultez le « [Informations de sécurité et de conformité Extron](#) » sur le site internet d'Extron.

Conforme aux section 1040.10 et 1040.11 du titre 21 du Code des règlements fédéraux des États-Unis.

Conventions Used in this Guide

Notifications

The following notifications are used in this guide:

CAUTION: Risk of minor personal injury.

ATTENTION : Risque de blessure mineure.

ATTENTION: Risk of property damage.

ATTENTION: Risque de dommages matériels.

NOTE: A note draws attention to important information.

Software Commands

Commands are written in the fonts shown here:

`^AR Merge Scene, 0p1 scene 1,1 ^B 51 ^W ^C.0`

`[01] R000400300004000080000600 [02] 35 [17] [03]`

`[Esc][X1]*[X17]*[X20]*[X23]*[X21]CE ←`

NOTE: For commands and examples of computer or device responses used in this guide, the character “0” is the number zero and “O” is the capital letter “o.”

Computer responses and directory paths that do not have variables are written in the font shown here:

Reply from 208.132.180.48: bytes=32 times=2ms TTL=32

C:\Program Files\Extron

Variables are written in *italics* as shown here:

ping xxx.xxx.xxx.xxx -t

SOH R *Data* STX *Command* ETB ETX

Selectable items, such as menu names, menu options, buttons, tabs, and field names are written in the font shown here:

From the **File** menu, select **New**.

Click the **OK** button.

Specifications Availability

Product specifications are available on the Extron website, www.extron.com.

Extron Glossary of Terms

A glossary of terms is available at <https://www.extron.com/technology/glossary.aspx>

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Introduction

This section provides an overview of the DSC 12G-HD A and DSC HD-12G A scalers, covering the following topics:

- [About This Guide](#)
- [About the DSC 12G-HD A and DSC HD-12G A](#)
- [Features](#)
- [Application Diagrams](#)

About This Guide

This guide describes the Extron DSC 12G SDI to HDMI and DSC HDMI to 12G SDI Digital Scaling Converters and provides instructions for experienced installers to install, configure, and operate them.

In this guide the terms scaler and DSC are used interchangeably to refer to both DSC models.

About the DSC 12G-HD A and DSC HD-12G A

The DSC 12G-HD A and DSC HD-12G A are single-input scalers that accept multiple video resolutions and offer multiple output rates.

- **The Extron DSC 12G-HD A** is a multi-format SDI to HDMI scaler. It converts 12G-SDI, 6G-SDI, 3G-SDI, and HD-SDI to HDMI, supporting SMPTE video resolutions from 720p through 4K @ 60 Hz, and provides selectable output rates up to 4K @ 60 Hz, along with custom output resolution capability. Audio from SDI or analog sources is embedded on the HDMI output and available on analog outputs. An SDI input loop is also included.
- **The Extron DSC HD-12G A** is an HDMI to multi-format SDI scaler with dual mirrored outputs. It converts HDMI video signals to 12G-SDI, 6G-SDI, 3G-SDI, and HD-SDI formats, supporting input resolutions from 480i through 4K @ 60 Hz, and provides selectable SDI output formats from 720p to 4K @ 60 Hz. Audio from HDMI or analog input sources is routed to both SDI and analog outputs. Additional features include genlock input with loop-through.

Both models incorporate the Extron-exclusive Vector™ 4K scaling engine for critical-quality 4K applications, motion-adaptive deinterlacing for signals up to 1080i, Deep Color processing, seamless upstream switching, and on-screen logo display for messaging and corporate branding. They provide 30-bit processing and 1080i deinterlacing as well as embedded audio. Other features of both models include Auto Memory, Image Reset, and internal test patterns.

Setup and control are provided via front panel controls with an on-screen display (OSD), a front panel USB CONFIG port, and rear panel LAN ports. SIS (Simple Instruction Set) commands and the Windows®-based Product Configuration Software (PCS) program can be used to configure and control the DSC via the USB CONFIG ports.

Both half-rack models have compact 1.125 inches (29 mm) high by 8.69 inches (222 mm) wide by 6.0 inches (152 mm) deep enclosures.

Features

DSC 12G-HD A

- **Multi-rate SDI to HDMI video scaling** — Provides high quality conversion of signal format and video resolution with audio embedding.
- **Accepts 12G-SDI, 6G-SDI, 3G-SDI, and HD-SDI signals at data rates from 1.48 Gbps to 11.88 Gbps** — Accepts video formats from 720p up to 4K @ 60 Hz.
- **SMPTE compliance** — Complies with SMPTE 292M, SMPTE 424M, SMPTE ST-2081, and SMPTE ST-2082 digital video standards.
- **Input** — One SDI on a BNC connector, and balanced and unbalanced stereo audio on a 5-pole captive screw connector.
- **Selectable output rates from 640x480 to 4K @ 60 Hz** — Available output rates include video resolutions from 640x480 to 4096x2160
- **Audio embedding** — Analog audio or up to four pairs of AES audio channels from the SDI input can be embedded onto the HDMI output.
- **Audio de-embedding** — Two channel LPCM SDI audio can be extracted to the analog audio output, or multi-channel and bitstream formats can be passed to the HDMI output.
- **Input equalization and reclocking** — Automatically equalizes incoming signals and reshapes and restores signals on the buffered loop-through.
- **Selectable output color space** — The color space can be set to RGB or component video.
- **HDMI to DVI interface format correction** — Automatically enables or disables embedded audio and InfoFrames, and sets the correct color space for proper connection to HDMI and DVI displays.

DSC HD-12G A

- **HDMI to 12G-SDI, HD-SDI, and SD-SDI video scaling** — Provides high quality conversion of signal format and video resolution with audio embedding.
- **Accepts HDMI signals up to 4K @ 60 Hz** — Accepts HDMI input signals at a variety of resolutions. Only unencrypted content can be sent to the output.
- **Outputs** — Two simultaneous 12G-SDI/6G-SDI/3G-SDI/HD-SDI dual multi-rate SDI outputs are available for driving two displays.
- **Inputs** — One HDMI connector, and balanced and unbalanced stereo audio on a 5-pole captive screw connector
- **Selectable output rates from 720p to 4K @ 60 Hz** — Available output rates include video resolutions from 1280x720 to 4096x2160.
- **Audio de-embedding** — Embedded two channel LPCM HDMI audio can be extracted to the analog output, or multi-channel and bitstream formats can be passed to the SDI output.
- **SMPTE compliance** — Outputs comply with SMPTE 292M, SMPTE 424M, SMPTE ST-2081, and SMPTE ST-2082 digital video standards.
- **EDID Emulation provides selectable resolutions and refresh rates to ensure reliable operation** — Provides a means for specifying the rate of the incoming signal, ensuring proper communication with the video source.
- **Genlock input with loop-out** — Allows for synchronization to an external reference signal and supports bi-level or tri-level sync for integration into broadcast and production applications.

Both Models

- **Advanced Extron Vector 4K scaling engine** — The Vector 4K scaling engine is specifically designed for critical-quality 4K imagery, with best-in-class image upscaling and downscaling.
- **Logo image keying and display** — A logo graphic can be positioned and keyed over live video. Logo graphics in BMP, JPG, PNG, or TIFF format may be uploaded to the unit. Full screen images up to 4K resolution can also be displayed to eliminate loss of video between presentations.
- **AFL - Accu-RATE Frame Lock** — A patented technology exclusive to Extron that locks the output frame rate to the input to eliminate stuttering caused by frame rate conversion.
- **Displays user-supplied images for screen saver, corporate branding, and logo insertion** — Custom user-loaded images can be displayed as a screen saver after a predefined duration of inactivity at the video input, or whenever the input is disconnected between presentations.
- **Image freeze control** — A live image can be frozen using Ethernet or USB.
- **Auto Input Memory** — When activated, the unit automatically stores size, position, and picture settings based on the incoming signal. When the same signal is detected again, these image settings are automatically recalled from memory.
- **Motion-adaptive deinterlacing for signals up to 1080i** — Advanced deinterlacing for all interlaced signals up to 1080i delivers optimized image quality.
- **Automatic 3:2 and 2:2 pulldown detection** — Advanced film mode processing techniques that help maximize image quality for 1080i sources that originated from film.
- **Automatic input cable equalization** — Actively conditions incoming HDMI signals to compensate for signal loss when using long cables, low quality cables, or source devices with poor signal output.
- **On-screen menus** — Intuitive on-screen menus allow for easy system setup using the front panel controls. Key parameters such as input and output video formats are conveniently grouped on the initial Quick Setup screen, while additional screens provide full control over the other functions and settings of the unit.
- **Picture controls for brightness, contrast, and detail, as well as horizontal and vertical positioning, and sizing.**
- **User presets** — Memory presets are available to store and recall optimized image settings.
- **Internal video test patterns and pink noise generator for calibration and setup** — The scaler offers several video test patterns and audio pink noise to facilitate proper system setup and calibration of display devices.
- **Audio input gain and attenuation** — Gain and attenuation can be adjusted for the analog stereo input.
- **Integrated audio delay** — The audio output is automatically delayed to compensate for latency introduced by the video processing.
- **Output volume control** — Provides master volume control for the analog and SDI embedded audio outputs.
- **Selectable audio muting** — The SDI-embedded and analog audio outputs can be independently muted.
- **Output Standby Mode** — The unit can be set to automatically mute video and sync output to the display device when no active input signal is detected. This allows the projector or flat-panel display to automatically enter into standby mode to save energy and enhance lamp or panel life.
- **Power Save Mode** — Can be placed in a low power standby state to conserve energy when not in use.
- **Meets Ecoefficient Level 1** — Includes an energy-saving, externally controllable standby mode that puts the device in a low power state when not in use.
- **Front panel security lockout** — Locks out all front panel functions. All functions, however, are available through Ethernet and USB.
- **Ethernet monitoring and control** — Enables control and proactive monitoring over a LAN or WAN. An intuitive Web interface is included for system monitoring and firmware updates.
- **Built in web pages** — Enables the use of a standard browser for monitoring, troubleshooting, and firmware updates over an intuitive Web interface.

- **Front panel USB-C configuration port** — Enables configuration without having to access the rear panel.
- **Front panel LED indicators for signal format, presence, and power** — Provides visual indication of system status for real-time feedback and monitoring of key performance parameters.
- **Easy setup and commissioning with Extron Product Configuration Software (PCS)** — Configure multiple products using a single software application.
- **1.125-inches high, half-rack wide enclosure.**
- **Includes a LockIt Plus cable lacing bracket** — Securely fastens USB-C or HDMI cables to a device, eliminating risk of disconnection.
- **Internal Extron Everlast power supply** — Provides worldwide power compatibility, with high demonstrated reliability and low power consumption for reduced operating cost.
- **Extron Everlast Power Supply is covered by a 7-year parts and labor warranty.**

Application Diagrams

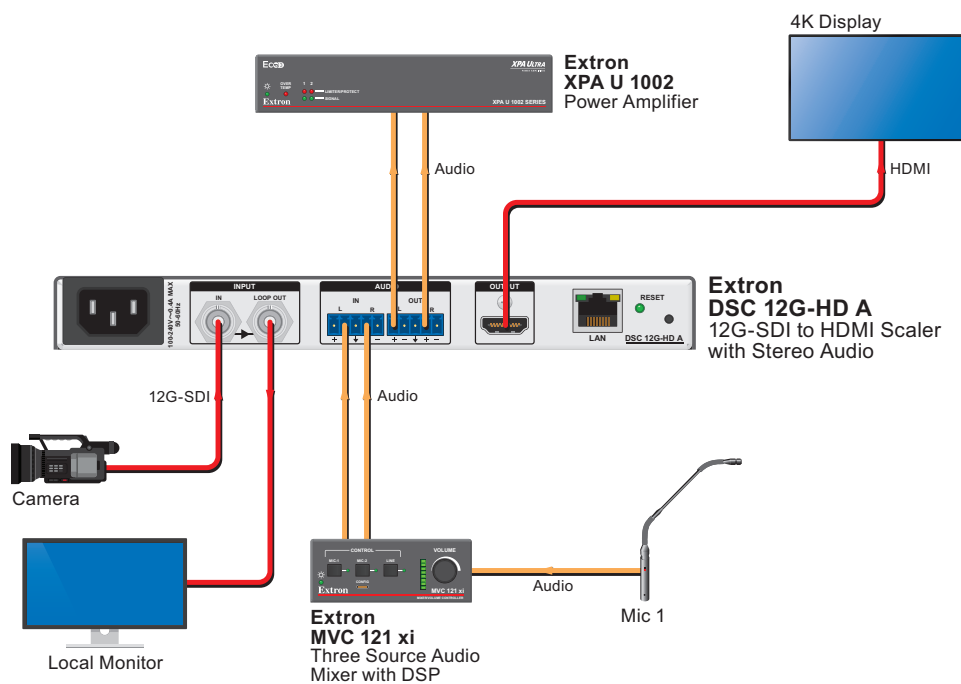


Figure 1. DSC 12G-HD A Application Example

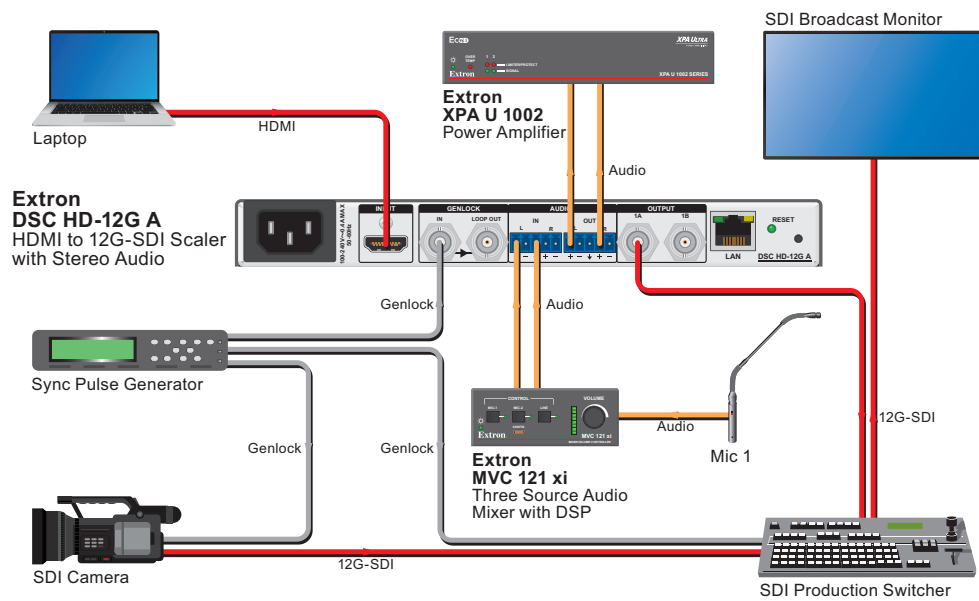


Figure 2. DSC HD-12G A Application Example

Installation

This section provides a description of the DSC 12G-HD A and DSC HD-12G A rear panel connectors and instructions for cabling. Topics include:

- [Rear Panel Connections](#)
- [LockIt Plus HDMI Cable Lacing Bracket Installation](#)
- [Connecting to the USB Config Port](#)

ATTENTION:

- Installation and service must be performed by authorized personnel only.
- L'installation et l'entretien doivent être effectués par le personnel autorisé uniquement.

Rear Panel Connections

Figures 3 and 4 show the rear panel features of the DSC 12G-HD A and DSC HD-12G A, respectively.

CAUTION: Remove power from the system before making any connections.

ATTENTION: Mettez le système hors tension avant d'effectuer tout raccordement.

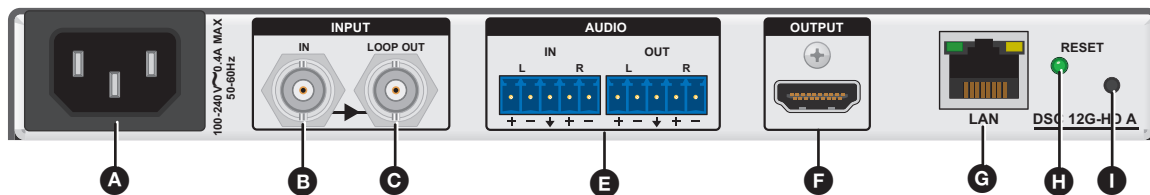


Figure 3. DSC 12G-HD A Rear Panel

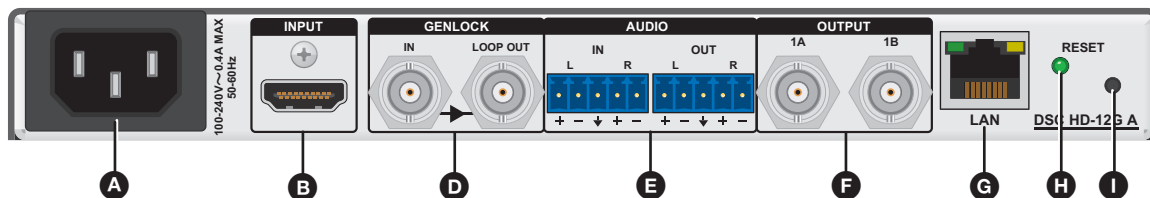


Figure 4. DSC HD-12G A Rear Panel

- | | |
|--|----------------------------|
| A AC power connector | F Output connectors |
| B Input connector | G LAN port |
| C LOOP OUT input connector (DSC 12G-HD A only) | H Reset LED |
| D GENLOCK and LOOP OUT connectors (DSC HD-12G A only) | I Reset button |
| E AUDIO connectors | |

ATTENTION:

- Use electrostatic discharge precautions (be electrically grounded) when making connections. Electrostatic discharge (ESD) can damage equipment, although you may not feel, see, or hear it.
- Prenez des précautions contre les décharges électrostatiques (soyez électriquement relié à la terre) lorsque vous effectuez des connexions. Les décharges électrostatiques (ESD) peuvent endommager l'équipement, même si vous ne pouvez pas le sentir, le voir ou l'entendre.

- A AC power connector** — Connect the supplied US standard IEC power cable between this IEC connector and a 110-220 V 50-60 Hz AC power source. The front panel control and input selection buttons light in sequence during power-up.

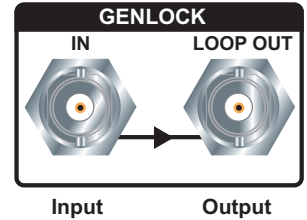
B Input connector —

- DSC 12G-HD A** — Connect an SDI video source to this 12G/6G/3G/HD-SDI BNC input connector. The DSC 12G-HD A accepts standard SDI input rates of 720p through 4K at 60 Hz.
- DSC HD-12G A** — Connect an HDMI source to this HDMI input connector, and secure it using the LockIt Plus bracket (see [LockIt Plus HDMI Cable Lacing Bracket Installation](#) on page 9). The DSC HD-12G A supports standard input rates of 480i/576i through 4K at 60 Hz.

- C LOOP OUT input connector (DSC 12G-HD only)** — Connect a local monitor to this SDI BNC loop out port.

- D GENLOCK and LOOP OUT connectors (DSC HD-12G A only)** — The BNC input connector (left, labeled GENLOCK) syncs the video signal in broadcast or other sync critical applications. The BNC output connector (right, labeled LOOP OUT) routes the sync signal throughout the system in broadcast or other sync-critical applications. Bi-level and tri-level sync are supported.

Connect an external sync device to the GENLOCK input connector, and connect any downstream equipment that requires genlocking LOOP OUT connector.



E AUDIO connectors —

- Audio Input** — This single 5-pole captive screw analog audio connector supports stereo balanced and unbalanced analog audio line input, and can be associated with the video input.

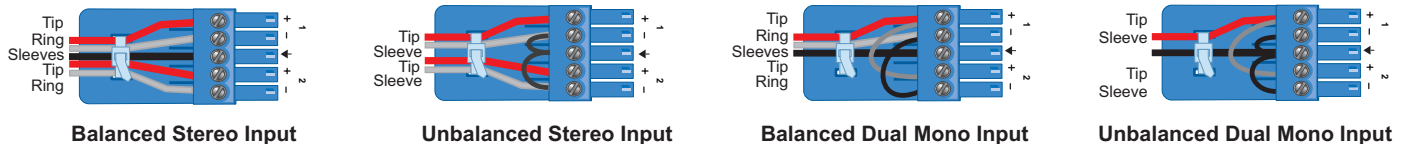


Figure 5. Wiring the Audio Input Connector

- DSC 12G-HD A** — Connect audio from the source to this 5-pole captive screw connector. Analog audio input can be embedded on the HDMI output. AES audio that is embedded on the SDI input can be identified and one to four pairs (8 channels) can be embedded on the HDMI output.

If eight-channel digital is selected, four of the eight available channel pairs are used.

If two-channel digital is selected, one of the eight pairs can be selected.

- DSC HD-12G A** — Connect analog audio input from the source to this 5-pole captive screw connector. HDMI or analog audio can be passed through to the SDI outputs as AES pairs.
- Audio Output** — This female five-pole captive screw connector supports stereo and mono balanced and unbalanced analog audio output. It passes program audio configured for use with the video input, and can either contain de-embedded audio from the source (2-channel LPCM only), or pass the analog input.

NOTE: The analog output and the embedded digital audio outputs (LPCM-2Ch only) are both affected by volume adjustment controls (Multi-Ch bitstream audio is not affected). If a Multi-Ch input is present, such as Dolby/DTS or multi-channel PCM audio, the analog output is muted.

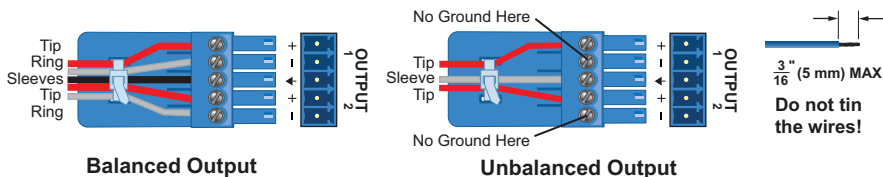


Figure 6. Wiring the Audio Output Connectors

Connect to an analog audio device such as an amplifier or DSP processor.

ATTENTION:

- The length of the exposed wires in the stripping process is critical. The ideal length is 3/16 inch (5 mm). If the exposed portion is longer, the wires may touch, causing a short circuit between them. If the exposed wires are shorter, they can be easily pulled out, even if tightly fastened by the captive screws.
- La longueur des câbles exposés est primordiale lorsque l'on entreprend de les dénuder. La longueur idéale est de 5 mm (3/16 inches). S'ils sont un peu plus longs, les câbles exposés pourraient se toucher et provoquer un court circuit. S'ils sont un peu plus courts, ils pourraient sortir, même s'ils sont attachés par les vis captives.
- Do not tin the wires. Tinned wires are not secure in the captive screw terminals and could pull out.
- Ne pas étamer les câbles. Les câbles étamés ne sont pas aussi bien fixés dans les terminaisons des à vis captives et pourraient sortir.
- For unbalanced audio, connect the sleeves to the ground contact. **Do not** connect them to negative (–) contacts.
- Pour l'audio asymétrique, connectez les manchons au contact au sol. Ne PAS connecter les manchons aux contacts négatifs (–).

F Output connectors —

- **DSC 12G-HD A HDMI output** — Connect an HDMI display device to this female HDMI connector, and secure it using the LockIt Plus bracket (see [LockIt Plus HDMI Cable Lacing Bracket Installation](#) on page 9).
- **DSC HD-12G A SDI output** — Connect one or two SDI output devices to these BNC connectors. These ports transmit signals that follow SMPTE standards 259M, 292M, and 424M.

- **LAN port** — To control the DSC through Ethernet, connect this RJ-45 connector to a network. Ethernet control allows configuration and control of the DSC from a remote location using SIS commands, PCS, or the embedded web pages.

When connected to an Ethernet LAN, the DSC series can be accessed from a computer running a standard Internet browser. Use a patch or crossover cable to connect the DSC to a computer, control device, router, or switch. The default IP address of the scaler is 192.168.254.254, the default subnet mask is 255.255.255.0, and the default gateway address is 0.0.0.0.

The LAN connector contains two LEDs (see the illustration at right):

- **Act LED** — This amber LED blinks to indicate LAN signal activity.
- **Link LED** — This green LED lights steadily to indicate a LAN connection.

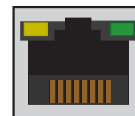


Figure 7 shows how to wire the LAN connector.

Pin	T568A Wire Color	T568B Wire Color
1	White-green	White-orange
2	Green	Orange
3	White-orange	White-green
4	Blue	Blue
5	White-blue	White-blue
6	Orange	Green
7	White-brown	White-brown
8	Brown	Brown

Pins:
1 2 3 4 5 6 7 8

Insert Twisted Pair Wires

RJ-45 Connector

Figure 7. Connecting to the LAN Port

- H Reset LED** — This green LED remains lit while the unit has power. While the **Reset** button (see [figures 3 and 4](#), **I** on page 6) is being pressed and held, this LED blinks every 3 seconds to indicate the level of reset initiated if the button is released at that point (see [Resetting](#) starting on page 32 for more information).
- I Reset button** — This recessed button initiates levels of reset on the DSC. To initiate the different reset types, use a pointed object such as an Extron Tweeker (or other small Philips screwdriver) or a stylus to press and hold the button while the scaler is running or while it is being powered up (see [Resetting](#)).

LockIt Plus HDMI Cable Lacing Bracket Installation

The Extron LockIt Plus lacing bracket secures a standard HDMI cable to most HDMI devices.

NOTE: The HDMI device must have an HDMI connection mounting screw for this bracket to be used.

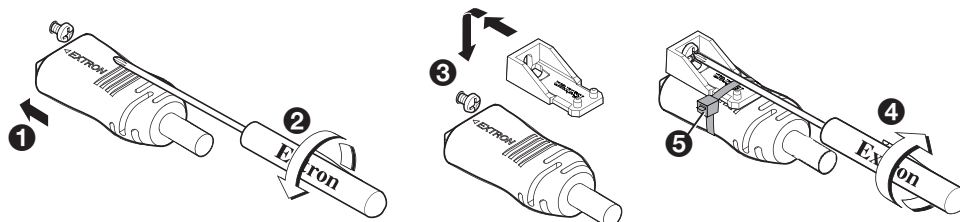


Figure 8. Mounting a LockIt Plus Cable Lacing Bracket

1. Plug the HDMI cable into the panel connection (**1**).
2. Loosen the HDMI connection mounting screw from the panel enough to allow the LockIt Plus lacing bracket to be placed over it (**2**).
3. Place the LockIt Plus lacing bracket onto the screw and slide it up against the HDMI connector. Tighten the screw to secure the bracket (**3**).
4. Loosely place the included tie wrap around the HDMI connector and LockIt Plus lacing bracket (**4**).
5. While holding the connector securely against the lacing bracket, tighten the tie wrap, then remove any excess length (**5**).

ATTENTION:

- Do not overtighten the HDMI connection mounting screw. The shield to which it is fastened is very thin and can easily be stripped.
- Ne serrez pas trop la vis de montage du connecteur HDMI. Le blindage auquel elle est attachée est très fin et peut facilement être dénudé.

Connecting to the USB Config Port

The USB-C CONFIG port is located on the DSC front panel. Use a USB cable to connect the DSC CONFIG port to a USB port on the computer.

Operation

This section discusses the functions available through the front panel and the on-screen display (OSD) to configure and operate the DSC. Topics include:

- [Front Panel Features](#)
- [On-screen Display](#)
- [Image Reset](#)
- [Changing the Output Resolution and Refresh Rate](#)
- [Power Save Mode](#)
- [Presets](#)
- [Locking the Front Panel \(Executive Mode\)](#)
- [Resetting](#)

Front Panel Features

Figure 9 and figure 10 show the front panel features of the DSC 12G-HD A and DSC HD-12G A, respectively.

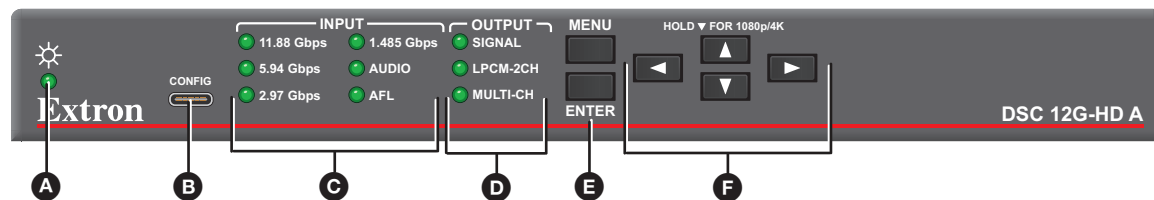


Figure 9. DSC 12G-HD A Front Panel

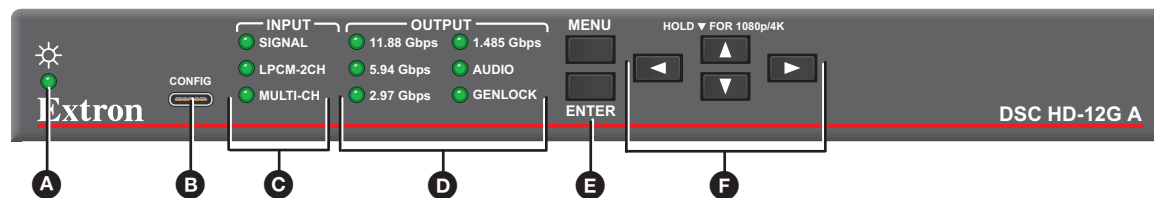


Figure 10. DSC HD-12G A Front Panel

- | | |
|----------------------|---------------------------------|
| A Power LED | D OUTPUT LEDs |
| B CONFIG port | E MENU and ENTER buttons |
| C INPUT LEDs | F Navigation buttons |

A Power LED —

- Lights steadily amber when the unit is powered on but no signal is present.
- Blinks amber every 3 seconds when the unit is in standby mode.
- Lights steadily green when both power and an input signal are present.

- B** **CONFIG port** — Connect a control system or computer to this female USB-C port for remote control of the scaler, including device configuration, file transfers, and firmware updates. Connect it to a host computer for configuration using SIS commands, PCS, Toolbelt, or the internal web pages via an SSH client and IP address 203.0.113.22 (SIS via SSH port 22023). This IP address is hard-coded and cannot be changed.

C INPUT LEDs (see [figures 9 and 10](#) on page 10)

• **DSC 12G-HD A INPUT LEDs** —

- One signal LED lights to indicate the detected SDI input format.
 - 11.88 Gbps — SMPTE ST 2082
 - 5.94 Gbps — SMPTE ST 2081
 - 2.97 Gbps — SMPTE 424M (level A only)
 - 1.485 Gbps — SMPTE 292M
- The AUDIO LED lights when audio is present on the SDI input.
- The AFL LED lights when HDMI output refresh rate is locked to the applied SDI input.
- **DSC HD-12G A INPUT LEDs** — These three green LEDs light to indicate presence of a video input signal and the type of audio signal if one is also present.
 - **SIGNAL** — Lights when an input signal is present.
 - **LPCM-2CH** — Lights when 2-channel audio is input over HDMI.
 - **MULTI-CH** — Lights when multi-channel audio is input over HDMI.

D OUTPUT LEDs

- **DSC 12G-HD A OUTPUT LEDs** — These four green LEDs light to indicate the presence of sync and other information about the signal being output.
 - **SIGNAL** — Lights when an output signal is present.
 - **LPCM-2CH** — Lights when 2-channel audio is being output over HDMI.
 - **MULTI-CH** — Lights when multi-channel audio is being output over HDMI.
- **DSC HD-12G A OUTPUT LEDs** — These four green LEDs light to indicate the presence of sync and the output rate specified for the output (output rates can be specified using SIS commands, PCS, or the OSD).
 - One signal LED lights to indicate the detected HDMI output format:
 - 11.88 Gbps — SMPTE ST 2082
 - 5.94 Gbps — SMPTE ST 2081
 - 2.97 Gbps — SMPTE 424M (level A only)
 - 1.485 Gbps — SMPTE 292M
 - The AUDIO LED lights when audio is embedded in the SDI signal.
 - The GENLOCK LED:
 - Lights steadily if genlock is enabled and the DSC is locked.
 - Blinks if genlock is enabled and there is no genlock input, or if a genlock input is present but the scaler is unable to lock.

E MENU and ENTER buttons — Press these buttons to access the OSD and to select menus and menu items (see [Using the Front Panel Buttons with the OSD Menus](#) on page 13 for detailed explanations of these buttons and the arrow buttons).

F Navigation buttons — Press these right ►, left ◄, up ▲, and down ▼ arrow buttons to step through the OSD menus and reset the output rate (hold ▼ for 5 seconds to toggle between 1080p at 60 Hz and 4K at 60 Hz).

On-screen Display

The on-screen display (OSD) enables the user to configure and adjust the DSC from menus displayed on a monitor or other display device connected to a rear panel output.

NOTE: The settings available through these menus can also be selected via SIS commands (see [Remote Configuration and Control](#) starting on page 34).

Menu Overview

The OSD menu system consists of seven menus and an information screen. To access the menus, press the front panel **MENU** button. With the main menu displayed, use the arrow buttons to step through the menus and menu items. Press the **ENTER** button to select highlighted items. The **DEVICE INFO** screen is read-only and does not provide selections.

The DSC 12G-HD A and [DSC HD-12G A](#) OSD menu tables below and on the next page show the menus and their items for each DSC.

The menu screen always displays the following items, regardless of which menu is displayed.

- **Company name and product name** are in the upper-left corner.
- **Current input resolution and refresh rate** are in the lower-left corner.
- **Current output resolution and refresh rate** are in the lower-right corner.

OSD Menu Summary Tables

The following tables summarize the menus and their options on the DSC 12G-HD A and the DSC HD-12G A.

OSD Menu — DSC 12G-HD A							
Menus	Menu Items						
DEVICE INFO (Read-only)	Unit Name	Firmware (Version and build)	Temp	Input: • Format • Signal	Output: • Format • Signal • AFL		
QUICK SETUP	Input Type	Output Rate	Audio Mute	Test Pattern	DHCP Mode	IP Address	
PICTURE CONTROLS	Image Reset	Image Position (H and V)	Image Size (H and V)				
INPUT	Input Type	Active Video	Capture EDID				
OUTPUT	Output Rate	Format	AFL	User Logo	Logo Position		
AUDIO	Audio Mute	Audio Format	2CH Stereo Source	Swap Group 1 Pair 1	Analog Gain/Att.	Audio Output Format	Audio Output Volume
ADVANCED	Test Pattern	Screen Saver	Screen Saver Timeout	Aspect Ratio	Auto Memory	Upstream Switch Effect	Factory Reset
COMMUNICATIONS	MAC Address	DHCP	IP Address	Subnet Mask	Gateway	USB Config IP	

OSD Menu — DSC HD-12G A							
Menu	Menu Items						
DEVICE INFO (Read-only)	Unit Name	Firmware (Version and build)	Temp	Input <i>n</i> : • Format • Signal	Output <i>n</i> : • Format	Genlock/AFL	
QUICK SETUP	Input Type	Input EDID	Output Rate	Auto Mute	Test Pattern	DHCP Mode	IP Address
PICTURE CONTROLS	Image Reset	Image Position (H and V)	Image Size				
INPUT	Input Type	Active Video	EDID				
OUTPUT	Output Rate	Format	Genlock/AFL	SDI Genlock Offset	User Logo	Logo Position	
AUDIO	Audio Mute	Audio Format	Swap Group 1 Pair 1	Analog Gain/Att	Audio Output Format	Audio Output Volume	
ADVANCED	Test Pattern	Screen Saver	Screen Saver Timeout	Aspect Ratio	Auto Memory (Status)	Upstream Switch Effect	Factory Reset
COMMUNICATIONS	MAC Address	DHCP	IP Address	Subnet Mask	Gateway	USB Config IP	

Using the Front Panel Buttons with the OSD Menus

The front panel navigation buttons enable you to access the OSD and make selections from its menus.

- **MENU** — Press this button to open and close the OSD. Pressing it from within a menu or a menu item returns you to the next higher menu level.

For example, if a menu item is selected in error, pressing **MENU** changes the highlighting of the item to an outline. From there, use the up and down arrow buttons to move the outline to the desired item.

- **ENTER** — Press this button to access the items on the displayed menu (moves a blue outline to the first menu item in the right column). Press this button again to select the outlined menu item. The **ENTER** button also opens the OSD, but does not close it.
- **Left arrow** ◀ — Press this button to adjust settings on screens displayed as a result of selecting menu items. For COMMUNICATIONS menu items (for example, **IP Address** and **Subnet Mask**) press this button to select the octet on the **left**, then use the ▲ and ▼ buttons to adjust the octet values.
- **Right arrow** ▶ — Press this button adjust settings on screens displayed as a result of selecting menu items. For COMMUNICATIONS menu items, press this button to select the octet on the **right**, then use the ▲ and ▼ buttons to adjust the octet values.
- **Up arrow** ▲ — Press this button to move upward through main menu and menu items. This button also adjusts settings on screens displayed as a result of selecting menu items.
- **Down arrow** ▼ — Press this button to move downward through main menu and menu items. This button also adjusts settings on screens displayed as a result of selecting menu items.

This button can also be used to reset the output rate (see [Changing the Output Resolution and Refresh Rate](#) on page 30 and in conjunction with the **MENU** button to lock the front panel (see [Locking the Front Panel \(Executive Mode\)](#) on page 32).

Using the OSD Menus

To access and use the OSD menus:

1. Press the **MENU** button once to open the main menu on the display. The menu opens with the **QUICK SETUP** menu displayed and the menu name highlighted (see figure 11).

Extron	DSC HD-12G A	
DEVICE INFO	Input Type	No Signal
QUICK SETUP	Input EDID	Auto
PICTURE CONTROLS	Output Rate	3840x2160 @ 60 Hz
INPUT	Audio Mute	Off
OUTPUT	Test Pattern	Color Bars
AUDIO	DHCP Mode	On
ADVANCED	IP Address	192 . 168 . 254 . 254
COMMUNICATIONS		
Input 1	Scaled Output	
No Signal	3840x2160 @ 60 Hz	

Figure 11. OSD Main Menu, Initial View — QUICK SETUP Menu (DSC HD-12G A Example)

2. To view a different menu, press the ▼ or ▲ button to move to it.
3. Press the **ENTER** button to access the menu items. The first item is outlined.
4. Press the ▼ or ▲ button to move the outline to the desired item.
5. Press the **ENTER** button to select the outlined menu item. The selected item is highlighted.

When a **PICTURE CONTROLS** menu item is selected, the OSD shrinks to display only the settings currently being adjusted (see figure 12).

Image Size	◆	H: *1920 V: *1080	◆
------------	---	-------------------	---

Figure 12. Example of a Selected Picture Controls Submenu Item

6. Press the ◀ or ▶ button to adjust the value as desired.
For **PICTURE CONTROLS** menu items:
 - a. Press the right or left arrow button to select the sub-item on the right or the left.
 - b. Use the ▲ and ▼ buttons to adjust the settings of the selected sub-item.
 - c. Repeat steps 6a and 6b for the other sub-item.
7. Press **ENTER** to confirm the new values. Press it again if action confirmation is required. The highlighting is replaced by the outline on the menu screen.
8. Do any of the following:
 - Press the ◀ or ▶ button to move to another item on the menu and repeat steps 5 through 7.
 - Press **MENU** to exit the menu.
 - Press **MENU** twice to exit the menu and close the main menu.

NOTES:

- In the following sections, the screen examples are from the DSC 12G-HD A, unless the DSC 12G-HD A and the DSC HD-12G A menus have different items (for example, the QUICK SETUP menus). In those cases, the menu screens from both products are shown.
- Values in menus that do not apply to the current input are shown as N/A.
- An asterisk following an item indicates that it is the default.
- The OSD times out and closes after 1 minute if no buttons are pressed.

QUICK SETUP Menu

This menu (see figure 12) contains a selection of menu items that are also available on other OSD menus. It allows you to make several basic configuration adjustments to the DSC from the same menu screen.

NOTE: On this menu, the DSC HD-12G A has an **Input EDID** item, which is not on the DSC 12G-HD A QUICK SETUP menu.

Extron	DSC 12G-HD A
DEVICE INFO	Input Type 12G-SDI
QUICK SETUP	Output Rate 3840x2160 @ 60 Hz
PICTURE CONTROLS	Audio Mute Off
INPUT	Test Pattern Off
OUTPUT	DHCP Mode On
AUDIO	IP Address 192 . 168 . 254 . 254
ADVANCED	
COMMUNICATIONS	
Input 1 3840x2160 @ 60.00Hz	Scaled Output 3840x2160 @ 60 Hz

Extron	DSC HD-12G A
DEVICE INFO	Input Type No Signal
QUICK SETUP	Input EDID Auto
PICTURE CONTROLS	Output Rate 3840x2160 @ 60 Hz
INPUT	Audio Mute Off
OUTPUT	Test Pattern Color Bars
AUDIO	DHCP Mode On
ADVANCED	IP Address 192 . 168 . 254 . 254
COMMUNICATIONS	
Input 1 No Signal	Scaled Output 3840x2160 @ 60 Hz

Figure 13. QUICK SETUP Menus

From the DSC 12G HD A or DSC HD 12G A QUICK SETUP menu, the following items can be configured:

- **Input Type** — This read-only field shows the type of input being received (12G-SDI or HDMI), or No Signal if no input is detected.
- **Input EDID** — (DSC HD-12G A only) Select this item to match the input EDID to the output rate or to set a discrete EDID (see the [Scaler resolution tables](#) starting on page 38 for EDID data). The default is **Auto** (matches the configured output resolution).
- **Output Rate** — This item lets the user set the resolution and refresh rate for the current output. The DSC 12G-HD A has 42 factory installed output resolutions and rates and 5 custom user defined blocks for new resolutions. The DSC HD-12G A has 30 factory output resolutions and rates (no custom rates) (see the [Scaler resolution tables](#)). The default output rate for both DSC models is 1080p @ 60 Hz.
- **Audio Mute** — Select this item to mute and unmute the output audio:
 - **Off** — Audio output is not muted (default).
 - **On** — Mutes both the analog and the embedded digital audio outputs.

- **Test Pattern** — This item lets you choose a test pattern to aid in setting up the DSC and the corresponding output display. Available patterns are: Off (no test pattern), Crop, Alternating Pixels, Crosshatch, Color Bars, Grayscale, and Audio Test (displays a crop pattern with OSD text AUDIO TEST and outputs 48 kHz, 24-bit pink noise).
- **DHCP Mode** — When DHCP is enabled (**On**), the unit obtains an IP address and other network settings from the DHCP server. The default is **Off**.
- **IP Address** — Press the ◀ and ▶ buttons to select an octet of the IP address. Press the ▲ and ▼ buttons to adjust the value of the selected octet. Repeat as needed. The default is 192.168.254.254.

PICTURE CONTROLS Menu

This menu lets you adjust the horizontal and vertical positions, and height and width of the image on the display, as well as perform an Image Reset. (The PICTURE CONTROLS menu is the same on both DSC models.)

Extron	DSC 12G-HD A	
DEVICE INFO	Image Reset	Press ENTER
QUICK SETUP	Image Position	H: 0000 V: 0000
PICTURE CONTROLS	Image Size	H: 3840 V: 2160
INPUT		
OUTPUT		
AUDIO		
ADVANCED		
COMMUNICATIONS		
Input 1	Scaled Output	
3840x2160 @ 60.00Hz	3840x2160 @ 60 Hz	

Figure 14. PICTURE CONTROLS Menu

The Picture Controls submenu adjusts picture settings.

NOTE: When the aspect ratio changes as a result of Fill or Follow commands being entered, the Image Position and Image Size values are updated accordingly.

- **Image Reset** — To restore the image on the active input to the default size and position, press the **Enter** button, then press ▶ to execute an Image Reset. Image Reset updates the horizontal and vertical image position, and the horizontal and vertical image size settings.

NOTE: The **Image Reset** submenu item is the same as the standard Image Reset Execute SIS command. However, there are other Image Reset options available through SIS commands (see [Image Reset](#) on page 30) or PCS (see the *DSC 12G-HD A and DSC HD-12G A Help File*). The options include to execute an Image Reset and fill the output and to execute an Image Reset and maintain the input aspect ratio. These commands ignore the current aspect mode setting, perform Image Reset on the input, and then select **Fill** or **Follow**.

- The following is performed during an Image Reset when the aspect ratio is set to **Fill**:
 - Horizontal and vertical image positions return to **0000**.
 - Horizontal and vertical image sizes match the current output resolution.

- The following is performed during an Image Reset when the aspect ratio is set to **Follow**:
- The horizontal and vertical image position and image size are set to maintain the native aspect of the input rate with regard to the current output resolution.

Set the aspect ratio on the ADVANCED menu (see [Aspect Ratio](#) on page 27).

- **Image Position** — Press the ◀ and ▶ buttons to select the horizontal (H) or vertical (V) position of the image. Press the ▲ and ▼ buttons to adjust the value of the selected position. The ranges are:
 - **Horizontal position:** -4096 through +4096 pixels. The default is 0000.
 - **Vertical position:** -2160 through +2160 lines. The default is 0000.
- **Image Size** — Press the ◀ and ▶ buttons to select the horizontal (H) or vertical (V) size of the image. Press the ▲ and ▼ buttons to adjust the value of the selected position.
 - **Horizontal size (width):** 10 through 8192 of signal.
 - **Vertical size (height):** 10 through 4320 of signal.

Adjusting the picture controls

When a Picture Controls menu item is selected, the OSD menu collapses so that the item is displayed alone on the screen to facilitate adjustment (see figure 15). The separate item field contains blue arrows that indicate which front panel arrow buttons to press to adjust the item.



Figure 15. Example of a Selected Picture Controls Submenu Item

After selecting the item to adjust, do the following:

1. With the separate item field displayed, use the navigation buttons to select and adjust the desired settings as indicated by the blue arrows on the screen. (To rapidly increment or decrement the values, press and hold the arrow button.)

Example: In figure 15, the blue right and left arrows before the H setting indicate that you can press the ◀ and ▶ buttons on the front panel to adjust the horizontal size. To adjust the vertical size, press the ▲ and ▼ buttons, as indicated by the blue up and down arrows after the V setting.

2. When finished, press **ENTER** to return to the OSD menu.

INPUT Menu

This menu allows you to perform certain input adjustments and to **view** the active, total, and start of lines and pixels. (The **Input EDID** item is available only on the DSC HD-12G A, while **Capture EDID** is available only on the DSC 12G-HD A.)

Extron	DSC 12G-HD A	Extron	DSC HD-12G A
DEVICE INFO	Input Type12G-SDI	DEVICE INFO	Input TypeHDMI
QUICK SETUP	Active VideoH: 3840 V: 2160	QUICK SETUP	Active VideoH: 1920 V: 1080
PICTURE CONTROLS	Capture EDIDPress ENTER	PICTURE CONTROLS	EDIDAuto
INPUT		INPUT	
OUTPUT		OUTPUT	
AUDIO		AUDIO	
ADVANCED		ADVANCED	
COMMUNICATIONS		COMMUNICATIONS	
Input 13840x2160 @ 60.00Hz	Scaled Output1080p @ 60 Hz	Input 11080p @ 60.00Hz	Scaled Output3840x2160 @ 60 Hz

Figure 16. INPUT Menus

- **Input Type** — This read-only field shows the type of input being received (12G/6G/3G/HD-SDI or HDMI), or No Signal if no input is detected.
- **Active Video** — Shows the number of pixels (H) and lines (V) in the active video area.
- **Capture EDID** (DSC 12G-HD A only) — This item allows the user to capture HDMI EDIDs from the output for use as a custom HDMI output resolution.

Five custom user-defined EDIDs are available via SIS and PCS. Each is set to a factory default of 1080p @ 60 Hz, but users can upload or capture custom EDID tables to each slot.

1. Select **Capture EDID**, then press **ENTER**. A list of the five custom EDID slots is displayed at the top of the OSD screen.

NOTES:

- If no EDID has been captured yet, the names displayed for the slots are the default: 1080p @ 60 Hz.
- If an EDID has been applied to a slot, last three letters of the EDID file name (in parentheses) represent the name of the display manufacturer (for example, EXN is an abbreviation of Extron).

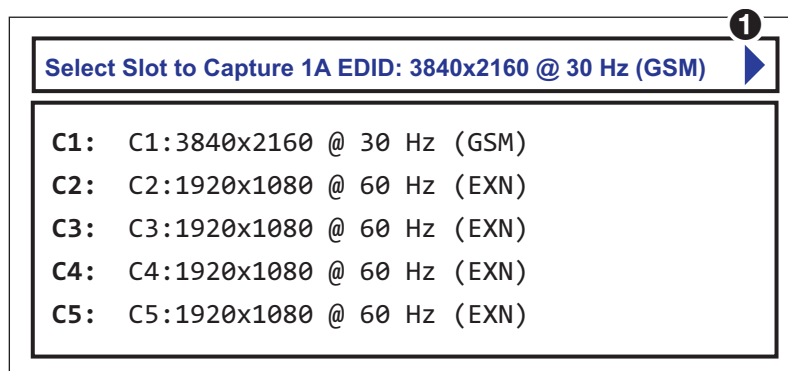


Figure 17. Custom EDID Slots List Example

2. Press the ◀ or ▶ button to select the display (output 1A or 1B) to assign the custom EDID. (The arrow button you press is reflected in the upper-right corner of the screen [see figure 17, ❶]).
3. Press the ▲ or ▼ button to highlight the slot to which you want to save the current display EDID.
4. Press **Enter**. The display EDID is saved to the selected custom slot and assigned to the connected inputs.

EDID (DSC HD-12G A only) — This item lets the user select an EDID (resolution and refresh rate) emulation for the HDMI input. The DSC HD-12G A products support standardized input rates of 480i/576i through 4K at 60 Hz. The HDMI input supports unique EDID emulation, deep color up to 12 bit, a 600 MHz pixel clock, and full 4K @ 60 Hz 4:4:4 video. Match the input EDID to the output rate or set a discrete EDID. The default is **Auto**, meaning the EDID matches the configured output resolution. The table [SIS Variables for DSC HD-12G A EDID](#) starting on page 39 shows the available EDID selections for the DSC HD-12G A.

Selecting an input EDID

To select a resolution and rate for input emulation:

1. From the **INPUT** menu, select **EDID**.
2. Press the ◀ or ▶ button to select a resolution.
3. Press the ▲ or ▼ button to select a rate for the resolution.
4. When finished, press **ENTER** to save the selection.

OUTPUT Menu

This menu allows you to configure the output of the DSC 12G-HD A or DSC HD-12G A.

Extron	DSC 12G-HD A	Extron	DSC HD-12G A
DEVICE INFO	Output Rate 3840x2160 @ 60 Hz	DEVICE INFO	Output Rate 3840x2160 @ 60 Hz
QUICK SETUP	Format Auto	QUICK SETUP	Format 12G-SDI YUV 422 Ltd
PICTURE CONTROLS	AFL Off	PICTURE CONTROLS	Genlock/AFL Off
INPUT	User Logo Off	INPUT	SDI Genlock Offset H: 0085 V: 0086
OUTPUT	Logo Position H: 0000 V: 0000	OUTPUT	User Logo Off
AUDIO		AUDIO	Logo Position H: 0000 V: 0000
ADVANCED		ADVANCED	
COMMUNICATIONS		COMMUNICATIONS	
Input 1 3840x2160 @ 60.00Hz	Scaled Output 3840x2160 @ 60 Hz	Input 1 1080p @ 60.00Hz	Scaled Output 3840x2160 @ 60 Hz

Figure 18. OUTPUT Menus

- **Output Rate** — This item lets the user select the resolution and refresh rate for the connected output display. Each DSC has a range of resolutions from which to choose. The available refresh rates depend on the selected resolution. See the following SIS variables tables for lists of available output rates for both models:
 - **DSC 12G-HD A** — See [SIS Variables for DSC 12G-HD A Resolutions and Refresh Rates](#) on page 38.
 - **DSC HD-12G A** — See [DSC HD-12G SIS Variables for EDID Resolution](#) starting on page 39.

To set the output resolution and refresh rate:

1. Select **Output Rate** from the OUTPUT menu.
2. Press ◀ or ▶ button to highlight the output resolution and then use up/down to set the desired output resolution.
3. If required, after setting the output resolution, press the ◀ or ▶ button to highlight the refresh rate, then press the ▲ or ▼ button to set the desired rate.
4. Press ENTER to save.

NOTE: If the resolution and rate selection are not confirmed within 15 seconds, the scaler returns to the previously selected resolution and rate.

- **Format** — This item lets the user set the HDMI output format (DSC 12G-HD A only).

To select the HDMI format, select **Format** from the OUTPUT menu, then press any of the arrow buttons to step through the items. The choices are:

DSC 12G-HD A

- **Auto** (based on the EDID of the sink) (default)
- **DVI RGB 444**
- **HDMI RGB 444 Full**
- **HDMI RGB 444 Ltd**
- **HDMI YUV 444 Ltd**
- **HDMI YUV 422 Ltd**

DSC HD-12G A (view only)

- **HD-SDI**
- **3G-SDI**
- **6G-SDI**
- **12G-SDI**

NOTE: The Format field on the DSC HD-12G A cannot be changed by the user. It reflects the SDI format of the connected scaler.

- **AFL (DSC 12G-HD A only)** — This item lets the user enable and disable the Extron Accu-RATE Frame Lock (AFL). The default is **Off** (disabled).

When enabled, this mode locks the output vertical refresh rate to the vertical refresh rate of the currently selected input using Accu-RATE Frame Lock technology. This ensures no frames of the input are repeated or dropped due to frame rate conversion. Input AFL mode can cause glitches or interruptions in the output sync when a new input is selected, or when a new signal has been routed to the selected input, as the DSC locks to the new vertical refresh rate.

NOTE: AFL locks the output refresh rate to similar input rates. If the input signal is locked on a rate that is not considered a standard HDMI output or that is very different from the current output setting, the DSC may not output a signal. The image may not display or may blink.

If no input signal is detected, or if locking to the input signal is not possible, a free running pixel clock is generated by the DSC.

Genlock/AFL (DSC HD-12G A only) —

- **AFL** (Accu RATE Frame Lock) locks the output vertical refresh rate to be to that of the applied HDMI input signal.
- **Genlock** locks the output vertical refresh rate to that of an applied reference sync signal.

NOTE: Genlock and AFL lock the output refresh rate to *similar* reference or input rates. If the input reference or the vertical refresh rate of the HDMI input signal do not adhere to SMPTE frequencies, the DSC may not output a signal, or the image may blink.

To select the genlock or AFL mode, select Genlock/AFL from the OUTPUT menu, then press any of the arrow buttons to step through the items. The choices are:

- **Off** — The output runs at the selected output frame rate using an internally generated clock. This selection is the default.
- **Input AFL** — The output is locked to the vertical sync of the HDMI input signal. This is valid only if HDMI input frame rate matches that of the DSC scaled output rate.
- **SDI Genlock** — The genlock input and loop-out are used for sync reference. The Genlock input supports bi-level and tri-level sync.

SDI Genlock locks the output vertical refresh rate to the applied analog genlock input on the DSC HD-12G A. In the SDI Genlock mode, the output resolution and refresh rate of the scaler must be set to exactly match the applied analog genlock signal to ensure a true genlock to the applied SDI Genlock signal.

If the applied SDI Genlock signal does not exactly match the resolution and refresh rate of the DSC HD-12G A output resolution, the scaler may be unable to lock, leading to a free running output clock or an unstable output.

If no SDI Genlock signal is detected, or if the detected SDI Genlock signal would result in a greater than 165 MHz pixel clock, the DSC generates a free running pixel clock.

The Genlock and AFL status can be queried via SIS command (see the [Genlock Offset](#) and [Input Lock \(Accu-Frame Lock \[AFL\]\)](#) commands on page 50) or via PCS (see the *DSC 12G-HD A and DSC HD-12G A Help File*).

NOTE: If the applied Genlock reference signal differs in resolution or refresh rate from the DSC scaled output resolution, then proper genlock cannot be guaranteed.

- **SDI Genlock Offset (DSC HD-12G A only)** — This item lets the user adjust the horizontal and vertical offset of the genlock signal. (This feature can be adjusted only if the DSC is set for SDI Genlock and actively locked to a reference input.)

Use the ◀ or ▶ button to adjust the horizontal (H) offset value. Use the ▲ or ▼ button to adjust the vertical (V) offset value.

- **User Logo** — Select this item to enable or disable display of a user-uploaded logo image. There are 16 slots available for storing logo files. To upload a logo file to the unit, use the Logo Config page of the PCS configuration program (see the *DSC 12G-HD A and DSC HD-12G A Help File* for the procedures).

To select a logo to display:

1. Select **User Logo** from the Output menu and press **ENTER**.
2. Press the ▲ or ▼ button to select a logo slot. By default the logo slots are named **01: Unassigned** through **16: Unassigned**. Use the Logos SIS commands (see the [Logos commands](#) on page 51) or the Logo Config screen of the PCS configuration program (see the *DSC 12G-HD A and DSC HD-12G A Help File*) to assign a logo file to the selected slot.

When a logo is assigned to a slot, its name changes from Unassigned to *nn:Logo n*. This name can be changed via SIS commands or PCS. The default selection for this menu is **Off**.

3. Press **Enter** to confirm your selection.

NOTE: To configure logo key effects (such as RGB keying, alpha keying, and so on) to key logos over live video see the *DSC 12G-HD A and DSC HD-12G A Help File*.

- **Logo Position** — This item lets the user specify a position for the selected logo on the display relative to the upper-left corner of the logo. The default is H: **0000** V: **0000**.

To position the logo:

1. Press the **ENTER** button to select **Logo Position** from the OUTPUT menu. The **Logo Position** item appears as a single field on the OSD.
2. Press the ▲ or ▼ buttons to select the horizontal position of the logo, and the ▶ or ◀ button to select the vertical position.
3. Press the **ENTER** button to confirm your selections.
4. Press ◀ to return to the OUTPUT menu.

AUDIO Menu

This menu allows the user to configure the audio that is processed through the DSC.

Extron	DSC 12G-HD A		Extron	DSC HD-12G A	
DEVICE INFO	Audio Mute	Off	DEVICE INFO	Audio Mute	Off
QUICK SETUP	Audio Format	LPCM-2Ch	QUICK SETUP	Audio Format	LPCM-2Ch
PICTURE CONTROLS	2CH Stereo source	Group 1 Pair 1	PICTURE CONTROLS	Swap Group 1 Pair 1	False
INPUT	Swap Group 1 Pair 1	False	INPUT	Analog Gain/Att	0 dB
OUTPUT	Analog Gain/Att	0 dB	OUTPUT	Audio Output Format	Stereo
AUDIO	Audio Output Format	Stereo	AUDIO	Audio Output Volume	0 dB
ADVANCED	Audio Output Volume	0 dB	ADVANCED		
COMMUNICATIONS			COMMUNICATIONS		
Input 1 3840x2160 @ 60.00Hz			Input 1 1080p @ 60.00Hz		
Scaled Output 3840x2160 @ 60 Hz			Scaled Output 3840x2160 @ 60 Hz		

Figure 19. AUDIO Menus

To use this menu, press **ENTER** to select an item, then press the **▲** or **▼** button to step through the selections.

- **Audio Mute** — Select this item to mute and unmute the output audio. Press any arrow key to select among **Off**, **On**, **Analog** (5-pole captive output), or **Digital** (embedded output).
- **Audio Format** — Select this item to choose the audio input format for the selected input. The first item on the AUDIO menu shows the selected format of the audio.

Audio Format	LPCM-2Ch
--------------	----------

Figure 20. Audio Format Field with Selected Audio Format

The audio format options are listed in the [table](#) on the next page.

Audio Format Selection	Details	Model
None	Mutes all audio input.	Both
Analog Audio	The audio from the 5-pole captive screw input port is embedded in the HDMI output.	DSC 12G-HD A
Analog Audio	The audio from the 5-pole captive screw input port is embedded in the SDI output.	DSC HD-12G A
LPCM-2Ch (default)	Lets the user select one pair of AES channels from the available eight pairs. If a format other than two-channel digital is selected for the audio format, the menu to select the pair is disabled (see Audio channels).	DSC 12G-HD A
LPCM-2Ch (default)	Configures the HDMI with LPCM-2Ch EDID and passes LPCM-2Ch audio to the SDI output.	DSC HD-12G A
Multi-Ch	Lets the user select up to four pairs (eight channels) of AES audio to be embedded in the HDMI output (see Audio channels).	DSC 12G-HD A
Multi-Ch	Configures the HDMI input with Multi-Ch EDID, and passes Multi-Ch audio to the SDI output (supports only LPCM sources).	DSC HD-12G A

- **2CH Stereo source (DSC 12G-HD A only)** — Allows for selection of which audio group or pair will be passed to the analog and embedded digital outputs (LPCM-2Ch mode only).
- **Swap Group 1 Pair 1** — Select **True** or **False** (default) to specify whether individual audio pairs will be swapped (Ch 1, Ch 2 versus Ch 2, Ch 1). This applies to the embedded digital output only.
- **Analog Gain / Att.** — Select this item to set the gain and attenuation for **analog** audio inputs **only**. The range is **-18 dB** to **+24 dB**. The decibel level can be adjusted in increments of 1.0 dB. The default is **0 dB**.
- **Audio Output Format** — Sets the audio output to **Stereo** (default) or dual **Mono** output mode (sums the left and right audio).
- **Output Volume** — Select this item to set the volume level in decibels for both the analog and embedded digital audio outputs. Press **Enter** to select the item, then press the **▲** or **▼** button to raise or lower the volume in increments of 1 dB. The range is **0 dB** through **-100 dB**. The default is **0 dB**.

Audio channels

On the DSC 12G-HD A the user can select channel pairs to embed, in addition to the audio input format. On the DSC HD-12G A, input format selection is available, but channel selection is not.

DSC 12G-HD A: Selecting audio channel pairs

The available formats (modes) for the DSC 12G-HD A are analog, two-channel digital, and multi-channel digital. In two-channel mode, one of the two channel pairs is embedded in the HDMI output. In four-channel mode, two pairs are embedded.

The eight channels are divided into two groups, each consisting of two channel pairs (4 channels). The table below shows the channels contained within each group.

Group 1				Group 2			
Ch Pair 1		Ch Pair 2		Ch Pair 1		Ch Pair 2	
Ch 1	Ch 2	Ch 3	Ch 4	Ch 5	Ch 6	Ch 7	Ch 8

- **Two-channel Digital Mode** — In this mode, the user can select a channel group, then one pair of AES channels from that group to embed, using either the SDI Audio Decode item of the Audio menu or SIS commands (see the [Audio Input Format commands](#) on page 53. To select a channel pair:

Press the and ► and ◀ or ▲ and ▼ buttons to scroll through **Group 1 Pair 1**, **Group 1 Pair 2**, **Group 2 Pair 1**, or **Group 2 Pair 2** to select the group number (1 or 2).

Multi-channel Digital (Full Digital) Mode — In this mode, the first four pairs (eight channels, groups 1 and 2) are embedded (see the table below). A channel pair cannot be selected.

Group	SDI Embedded Audio	HDMI Audio Signal
1	Pair 1 — Channel 1	L – Left speaker
	Pair 1 — Channel 2	R – Right speaker
	Pair 2 — Channel 1	LFE – Low frequency extension (woofer)
	Pair 2 — Channel 2	C – Center
2	Pair 1 — Channel 1	LS – Left surround
	Pair 1 — Channel 2	RS – Right surround
	Pair 2 — Channel 1	SBL – Surround back left
	Pair 2 — Channel 2	SBR – Surround back right

DSC HD-12G A: Audio embedding

In PCM multi-channel mode, the DSC HD-12G A embeds up to eight channels of PCM HDMI audio onto the first four pairs of AES channels on the SDI output, in sequential order. In LPCM two-channel mode, the first pair of SDI audio channels is used (the channel pair is not selectable on the DSC HD-12G A).

All audio inputs on the DSC HD-12G A are automatically delayed to compensate for internal video processing.

The table below shows the channel pair mapping for the DSC HD-12G A.

SDI Embedded Audio	HDMI Audio Signal
Group 1 Pair 1 – Channel 1	L – Left speaker
Group 1 Pair 1 – Channel 2	R – Right speaker
Group 1 Pair 2 – Channel 1	LFE – Low frequency extension (woofer)
Group 1 Pair 2 – Channel 2	C – Center
Group 2 Pair 3 – Channel 1	LS – Left surround
Group 2 Pair 3 – Channel 2	RS – Right surround
Group 2 Pair 4 – Channel 1	SBL – Surround back left
Group 2 Pair 4 – Channel 2	SBR – Surround back right

ADVANCED Menu

This menu allows the user to configure the global settings for the unit. The ADVANCED menu is the same on both DSC models.

Extron	DSC HD-12G A	
DEVICE INFO	Test Pattern	Color Bars
QUICK SETUP	Screen Saver	Black
PICTURE CONTROLS	Screen Saver Timeout	Never
INPUT	Aspect Ratio	Fill
OUTPUT	Auto Memory	On
AUDIO	Upstream Switch Effect	Low Latency
ADVANCED	Factory Reset	Hold ENTER
COMMUNICATIONS		
Input 1		Scaled Output
1080p @ 60.00Hz		3840x2160 @ 60 Hz

Figure 21. ADVANCED Menu, DSC HD-12G A Example

The following items are available on the ADVANCED menu:

- **Test Pattern** — Use this item to select a test pattern to use in setting up a display when outputting different resolutions. The following test patterns are available:

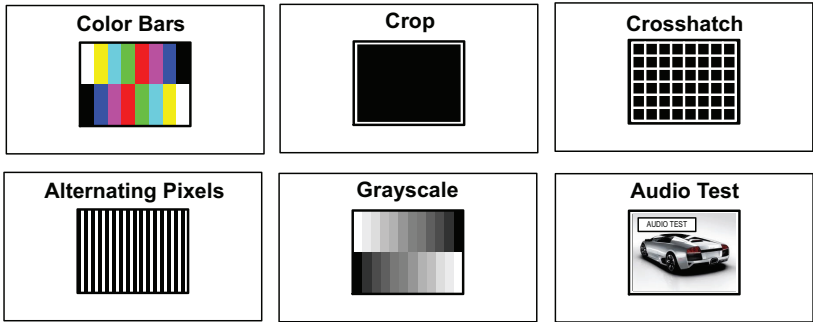


Figure 22. Test Pattern Items

NOTES:

- The Audio Test pattern displays a crop pattern with OSD text AUDIO TEST. It also outputs pink noise at 48 kHz, 24 bit.
- By default all test patterns include a single pixel wide crop pattern line.

- **Screen Saver** — Configures device behavior when no active input signal is detected. Options include a black (default) or blue screen, or a user provided logo file. Optionally, a timeout can be set to disable the output after a set duration to allow display devices (such as a display screen or projector) to enter a lower power or standby state to increase panel or lamp life.

To select the screen saver:

1. Select **Screen Saver** from the ADVANCED menu and press **ENTER**.
 2. Press the **▲** or **▼** button to select the screen saver type. The menu selections are:
 - **Black** (default)
 - **Blue** with the OSD message DSC {model name}: No Signal
 - **User Image** — Displays an image that has been previously uploaded to the DSC. The location of this image on the screen can be adjusted using SIS commands (see [Horizontal Shift - Logo](#) and [Vertical Shift - Logo](#) on page 51) or PCS (see the *DSC 12G-HD A and DSC HD-12G A Help File*).
 3. Press **ENTER** to confirm your selection.
- **Screen Saver Timeout** — Use this item to specify the number of seconds the selected screen saver is displayed before the output sync times out.

To set the sync timeout:

- Select **Screen Saver Timeout** from the ADVANCED menu and press **ENTER**.
- Press the **▲** or **▼** button to select the amount of time before output sync times out. The user can select a duration of **0** (timeout is immediate) to **500** seconds before the output sync times out. Select **NEVER** (the default) to set the sync to never time out.

NOTE: The DSC exits screen saver mode when the front panel **MENU** or **ENTER** button is pressed, a video test pattern is activated, or an active input is detected.

If the unit is in front panel lock mode (see [Front Panel Lock \(Executive Mode\) commands](#) on page 54), pressing a front panel button exits time out but access to any front panel controls or the menu system is disabled.

- **Aspect Ratio** — Sets the aspect ratio for the selected input to **Fill** (fills the entire output raster) or **Follow** (follows the native aspect ratio of the input).
 - **In Fill mode**, to set an aspect ratio adjustment for a single input rate, select the correct image size and position manually from the PICTURE CONTROLS menu (Image Size and Position) (see [PICTURE CONTROLS Menu](#) on page 16).
 - **In Follow mode**, each input rate is displayed with its native aspect ratio (4:3, 5:4, 16:9, or 16:10) with the correct letter box or pillar box settings, visible under the Image Size and Image Position items on the PICTURE CONTROLS menu.

If you want a single input to fill the screen in follow mode, you can manually set the Image Position item on the Picture Controls submenu to 0,0 and set the Image Size to match the current output rate X, Y.
 - **Auto Memory** — Enables or disables Auto Memory. When Auto Memory is set to On (default), the DSC stores the current input configuration and picture control values.
- The scaler stores 32 auto memories per input, with input configuration and picture control data for each video resolution. The default setting enables these memories to automatically recall input and picture controls for previously applied signals. When auto memories are disabled, the scaler treats every applied signal as a new source.
- **Upstream Switch Effect** — Enables selection of a transition effect to be used when the signal from an upstream switcher is lost and when it is restored. All fade switch effects have a duration of 0.5 seconds. Available effects are:
 - **Cut through Black** — The video mutes to black when the signal is lost and cuts to the new signal when it is restored.
 - **Fade through Black** — The last solid frame before the signal was lost freezes and the image fades to black. The new signal fades in when it is restored.

- **Seamless Fade** — The last solid frame before the signal was lost freezes, and a seamless dissolve to the new signal is performed when a new signal is applied. If no signal is applied for 8 seconds, the image on the display fades to black.
- **Seamless Cut** — The last solid frame before the signal was lost freezes, and the video seamlessly cuts to the new signal when the signal is restored. If no signal is applied for 8 seconds, the image on the display cuts to black.
- **Low Latency Mode** — The DSC attempts to display a new signal as quickly as possible. Two frames of video delay are also removed from the video processing logic of the DSC, which may introduce video artifacts during the switch transition (default).
- **Factory Reset** — This item lets the user reset the unit to its factory-installed values (removing any user-defined values) while retaining the current firmware version.

To reset using the OSD, press and hold the **Enter** button until **Factory Reset** is displayed. The message remains for approximately 1 minute after the reset is complete to allow time for the display device to sync with the DSC output. (The equivalent SIS command is **[Esc] ZXXX ←**.) For other reset methods, see [Resetting](#) starting on page 32.

COMMUNICATIONS Screen

The COMMUNICATIONS menu enables the user to view the unit media access code (MAC) hardware address and USB Config IP address, and to configure the DHCP mode, IP address, subnet mask, and gateway address.

Extron	DSC 12G HD A
DEVICE INFO	MAC Address 00-05-A6-2E-FD-7D
QUICK SETUP	DHCP On
PICTURE CONTROLS	IP Address 192 . 168 . 254 . 254
INPUT	Subnet Mask 255 . 255 . 255 . 0
OUTPUT	Gateway 0 . 0 . 0 . 0
AUDIO	USB Config IP 203 . 0 . 113 . 22
ADVANCED	
COMMUNICATIONS	
Input 1 3840x2160 @ 60.00Hz	Scaled Output 3840x2160 @ 60 Hz

Figure 23. COMMUNICATIONS Menu (DSC 12G-HD A Example)

- **MAC Address** — Displays the MAC address of the device (view only).
- **DHCP** — When DHCP is enabled (**On**), the unit obtains an IP address and other network settings from the DHCP server. The default is **Off**.

NOTE: When DHCP is enabled, all other IP fields become read-only.

To enable or disable DHCP mode:

1. Select **DHCP** from the COMMUNICATIONS menu and press **ENTER**.
2. Press the **▲** and **▼** button to select **On** or **Off** and press **ENTER** again.

The DHCP mode can also be set via PCS (see the *DSC 12G-HD A and DSC HD-12G A Help File*) or the internal web page (see [Network Panel](#) on page 65).

- **IP Address** — Press the ◀ and ▶ buttons to select an octet of the IP address. Press the ▲ and ▼ buttons to adjust the value of the selected octet. The default is 192.168.254.254.
- **Subnet Mask** — Press the ◀ and ▶ buttons to select an octet of the subnet mask address. Press the ▲ and ▼ buttons to adjust the value of the selected octet. The default is 255.255.255.0.
- **Gateway** — Press the ◀ and ▶ buttons to select an octet of the gateway address. Press the ▲ and ▼ buttons to adjust the value of the selected octet. The default is 0.0.0.0.
- **USB Config IP** — Displays the IP address (view only). The fixed value is 203.0.113.22.

Device Info Screen

This read-only screen provides a quick view of certain selections, parameters, and status information for the DSC. This screen differs between the two DSCs as follows:

- The DSC 12G-HD A DEVICE INFO screen (see) shows the AFL On and Off status and the display resolution and refresh rate (no genlock information).
- The DSC HD-12G A DEVICE INFO screen shows the Genlock/AFL On and Off status (no separate items for AFL or the display EDID).

Extron	DSC 12G-HD A	Extron	DSC HD-12G A
DEVICE INFO	Unit Name: DSC-12G-HD-A-2E-FD-7D Firmware: 1.00.0000-b006 Temp: 63.4C / 146.1F 11.6C Under Max Operating Temp	DEVICE INFO	Unit Name: DSC-HD-12G-A-30-75-B5 Firmware: 1.00.0001-b002-ENG_DEV Temp: 70.2C / 158.4F 4.8C Under Max Operating Temp
QUICK SETUP		QUICK SETUP	
PICTURE CONTROLS	Input: 3840x2160 @ 60.00 Hz Format: YUV 422 BT.709 Signal: 594.9 MHz 2250 Total Lines	PICTURE CONTROLS	Input 1: 1920x1080 @ 60.00 Hz Format: RGB 444 Limited Signal: 149.508 MHz 1125 Total Lines
INPUT	Output: 3840x2160 @ 60.00 Hz Format: HDMI RGB 444 Full Display: 3840x2160 @ 60 Hz (EXN) AFL: Off	INPUT	Output 1: 3840x2160 @ 60.00 Hz Format: 12G-SDI YUV 422 Limited
OUTPUT		OUTPUT	Genlock/AFL: Off
AUDIO		AUDIO	
ADVANCED		ADVANCED	
COMMUNICATIONS		COMMUNICATIONS	
Input 1 3840x2160 @ 60.00Hz	Scaled Output 3840x2160 @ 60 Hz	Input 1 1080p @ 60.00Hz	Scaled Output 3840x2160 @ 60 Hz

Figure 24. DEVICE INFO Screens

Image Reset

When Image Reset is performed, the DSC attempts to size and position the image to fill the display (fill mode) or to use the native aspect ratio of the input (follow mode). Using SIS commands or the OSD menus, you can perform a one-time Image Reset on the current input.

Only the horizontal and vertical positioning and the horizontal and vertical size are adjusted by the Image Reset. All other controls are left unchanged.

- If the aspect ratio has been set to **Fill**, the H and V position returns to 0,0, and the H and V size is set to match the current output rate.
- If the aspect ratio has been set to **Follow**, the H and V position and size are set to maintain the native aspect ratio of the input rate with respect to the current output resolution.

Image Reset is useful in applications in which a variety of new input sources may be encountered (such as when a matrix switcher is connected). It is also helpful in centering and sizing an image that has been adjusted out of default settings.

To perform a one-time Image Reset, use one of the following methods:

- **OSD menus** — Select the **Image Reset / Press Enter to Image Reset** item from the PICTURE CONTROLS menu (see [Image Reset](#) on page 16).
- **SIS commands** (see the [Image Reset commands](#) on page 44).
- **PCS** (see the *DSC 12G-HD A and DSC HD-12G A Help File*).

Changing the Output Resolution and Refresh Rate

In addition to the OUTPUT menus (see [OUTPUT Menu](#) starting on page 20), other options for setting the output resolution and refresh rate include resetting the output rate using the front panel ▼ button (see [Changing the Output Rate from the Front Panel](#)), SIS commands (see the [Output Scaler Rate commands](#) on page 48, and the PCS Configuration Program (see the program help file).

Custom Rates — DSC 12G-HD A Only

In addition to the 42 factory output resolutions, 5 custom user-defined rates are available for the DSC 12G-HD A via SIS commands (see the [Output Scaler Rate commands](#) for the command format) or PCS (see the program help file). When no custom EDIDs are defined, these five resolutions and rates default to 1080p @ 60 Hz. The OSD dynamically updates itself based on the current file stored in the custom EDID slot. For example, if a custom 1080p EDID has been uploaded, the OSD would read C1: 1080P @ 60 Hz.

Importing an HDMI EDID — DSC 12G-HD A Only

The EDID of a connected HDMI sink can also be imported to use as a custom rate. This EDID can be saved and stored in one of five custom EDID slots, which are then available for custom EDID emulation or custom output rate generation (custom rate based on the EDIDs preferred timings) (see the [Save Output EDID commands](#) on page 43).

NOTE: If an imported custom rate is not a valid output resolution, the output reverts to the default (1080p @ 60 Hz).

Changing the Output Rate from the Front Panel

To change the output rate from the front panel, press and hold the **DOWN** arrow navigation button for 5 seconds to switch output resolutions. Continued pressing of the **DOWN** arrow for an additional 5 seconds changes the output rate again in accordance with the chart on the next page. This adjusts the output resolution of the unit based on the current output resolution, which is useful during troubleshooting if the output display shows no video.

NOTE: This feature is NOT available when the unit is in executive mode or when the OSD menu system is in use.

When the down arrow has been pressed for the required duration of time, triggering an output resolution change, the front panel LED columns (OUTPUT LEDs only) blink twice all together for approximately 100 ms on and 100 ms off.

Rate at First Press of ▼ Button	After 5 seconds, changes to:
Rate other than 4K @ 60Hz or 1080p @ 60 Hz	1080p @ 60 Hz
1080p @ 60 Hz	4K @ 60 Hz
4K @ 60 Hz	1080p @ 60 Hz

Power Save Mode

In power save (standby) mode, all non-essential hardware is shut down in order to conserve power.

- To enable power save mode, enter the Power Save On SIS command (see the [Power Save Mode commands](#) on page 48).
- To exit power save mode, either enter the Power Save Off SIS command or press any front panel button. If the front panel is locked, this method does not work.

Presets

A preset is a set of input parameters that are either saved by the user or automatically by the DSC as a files in DSC memory. Saved presets can be recalled and their settings implemented at any time. A preset saves time by enabling the user to apply a group of settings to the DSC at one time.

The DSCs have two types of presets: Auto Memories and global input presets.

Auto Memories

The DSC has 32 memory locations in which the current input configuration and picture control values are automatically stored as an input preset, when Auto Memory is enabled. An Auto Memory preset is recalled when an identical input resolution and rate is applied to the input.

Each Auto Memory preset is stored with a time stamp that is incremented each time its stored rate is detected. If all Auto Memory slots are occupied and a new rate is applied, the Auto Memory with the oldest time stamp is overwritten.

Auto memory can be enabled using the OSD (see [Auto Memory](#) on page 27) and by SIS commands (see the [Auto Memories commands](#) on page 47).

The following information is stored in an Auto Memory preset:

• Horizontal start	• Vertical image position
• Vertical start	• Horizontal image size (width)
• Active pixels (Horizontal Active)	• Vertical image size (height)
• Active lines (Vertical Active)	• Total pixels
• Horizontal image position	

Input Presets

The DSC has 128 memory slots in which you can save input presets that are global to the HDMI or SDI input. These presets allow a matrix switcher with multiple types of video inputs (such as an Extron FOX, XTP, or DXP) to be placed upstream from the DSC to expand the number of video sources. Input presets can be saved and recalled only through SIS commands (see the [Input Presets commands](#) on page 47).

Values for the following settings are saved in input presets:

- Preset name
- Horizontal image position
- Vertical image position
- Horizontal image size
- Vertical image size

Saving a preset when using a matrix switcher

When using the DSC with a matrix switcher, do the following to save a preset:

1. Switch each input of the matrix to the input of the DSC.
2. Configure each input size and position.
3. Save the settings to a preset for recall by the control system when that matrix input is routed to the HDMI or SDI input.

Locking the Front Panel (Executive Mode)

To prevent access by unauthorized users or accidental changes to the DSC settings, the front panel controls can be locked by placing the DSC in executive mode. In this mode, all front panel buttons are disabled and control is available only by SIS commands.

To lock or unlock the front panel, enter the **Enable** or **Disable** SIS command (see the [Front Panel Lock \(Executive Mode\) commands](#) on page 54). Alternatively, select the executive mode via PCS (see the *DSC 12G-HD A and DSC HD-12G A Help File*).

Resetting

Resetting the unit causes various settings to revert to the defaults that were set at the factory. Resets can be performed from the front panel buttons, the OSD, or SIS commands.

The DSC 12G-HD A and HD-12G A scalars have four reset modes. Use a small screwdriver or stylus to press and hold the recessed **RESET** button on the rear panel to advance through the modes. The RESET LED blinks to indicate each mode that is enabled (see the Reset Mode Summary table for a description of the reset modes and how to select them).

Resets can also be performed using the OSD (see [Factory Reset](#) on page 28), SIS commands (see the [Resets commands](#) on page 54), or PCS (see the *DSC 12G-HD A and DSC HD-12G A Help File*).

Reset Mode Summary			
Mode	Activation	Result	Purpose and Notes
Use Factory Firmware	Using an Extron Tweezer or other small screwdriver, press and hold in the recessed RESET button for 30 seconds while applying power to the scaler. NOTE: After this reset, update the device with the latest firmware version. DO NOT operate with the firmware version that results from this type of reset. This temporarily resets the device to factory default until power is recycled. To use factory default firmware, upload that version again.	The device reverts to the factory default firmware. - Firmware reverts to the factory default for a single power cycle. - All user files and settings (drivers, audio and video adjustments, IP settings, and so on) are maintained. NOTE: If you do not want to update the firmware or if you perform a factory firmware reset by mistake, cycle power to the device to return to the firmware version running prior to the reset.	Use this mode to revert to the factory default firmware version for a single power cycle if incompatibility issues arise with user-loaded firmware.
Reset IP Settings	Hold in the RESET button until the RESET LED blinks twice (approximately 6 seconds). Then, press RESET momentarily (less than 1 second).	IP settings revert to factory defaults. - Port mapping reverts to factory default. - DHCP turns off. - The IP address is set to default. - The RESET LED blinks four times in quick succession during reset.	Use this mode to reset all IP settings back to factory defaults.
Reset to Factory Default	Hold down the RESET button until the RESET LED blinks three times (approximately 9 seconds). Then, press RESET momentarily (less than 1 second).	The device reverts to the factory defaults except for the firmware version. - All Reset IP Settings mode results are performed. - All user modifiable configurations are reset to default values including IP settings and real-time adjustments. - All user loaded files are deleted. - The RESET LED blinks four times in quick succession during the reset.	Use this mode to restart with default configuration. This is equivalent to the SIS command ZQQQ. This reset also removes the initial serial number password and sets it to extron.
DHCP Toggle	Quickly press the RESET button five times within 2 seconds.	- If the DHCP client is disabled, it is enabled. The RESET LED blinks six times. - If the DHCP client is enabled, it is disabled and the unit returns to the previous static IP address values. The RESET LED blinks three times.	Use this mode to turn the DHCP client on and off.

NOTE: The reset modes listed close all open IP and SSH connections and all sockets.

ATTENTION:

- Review the reset modes carefully. Using the wrong reset mode may result in unintended loss of flash memory programming, port reassignment, or processor reboot.
- Étudier de près les différents modes de réinitialisation. Appliquer le mauvais mode de réinitialisation peut causer une perte inattendue de la programmation de la mémoire flash, une reconfiguration des ports ou une réinitialisation du processeur.

NOTE: The Firmware Reset and Factory Reset messages are displayed for 1 minute after the reset to allow time for the display device to sync with the DSC output.

Remote Configuration and Control

This section describes the connections through which the DSC can be configured and controlled remotely via SIS commands and describes the commands that are available. It also provides the procedure for accessing the PCS configuration software. Topics include:

- [Host and Scaler Communication](#)
- [SIS Commands](#)
- [Product Configuration Software](#)

Host and Scaler Communication

SIS commands consist of one or more characters per field. No special characters are required to begin or end a command sequence. When the scaler determines that a command is valid, it executes the command and sends a response to the host device. All responses from the scaler to the host end with a carriage return and a line feed (CR/LF = ↵), indicating the end of the response character string (one or more characters).

Connecting the DSC to the host

Use one of the following methods to establish communication between the host and the DSC:

- **Ethernet** — Ensure the unit is connected to the host via the rear panel LAN port. To enter SIS commands, use a secure communication utility that supports Secure Shell (SSH). Enter the DSC IP address where requested and use **22023** as the port number. The default IP address is
- **Ethernet over USB** — Connect the host to the front panel USB Config port. To enter SIS commands, use a secure communication utility that supports Secure Shell (SSH). Enter **203.0.113.22** for the IP address where requested and **22023** as the port number.

To log in to the scaler via Ethernet or Ethernet over USB:

- **User name** = admin
- **Password** = {unit serial number} if the unit is new, extron if the unit has had an absolute system reset.

NOTE: Performing a unit factory reset (entering a ZQQQ [Absolute system reset](#)) or a [Use Factory Firmware](#) reset via the rear panel **Reset** button (see [page 33](#)) changes the serial number password to extron.

Password Messages

If the DSC is protected by a password, the following password message prompts you for the password to access the scaler features.

↵Password↵

The prompt requires a password, followed by a carriage return. The prompt is repeated if the correct password is not entered.

- **For initial setup:** On the rear panel of the DSC 12G-HD A and DSC HD-12G A, a label is attached containing the following text:

Pursuant to California Senate Bill 327, the password for all accounts on this device have been set to the device serial number. Passwords are case sensitive. See user guide for reset instructions.

This means that, for initial setup, the user must enter the unit serial number at the password prompt (the serial number can be found on another label on the rear panel). This password allows administrator level access.

If the correct password is entered, access is granted and the command prompt is displayed. If the password is incorrect, the `←Password←` prompt reappears.

- **After a password change:** After the initial access, you can keep the serial number password or change it. You can also set administrator and user level passwords (see the [Passwords commands](#) on page 56).

When you enter the new password correctly, the unit responds with one of the following, depending on the password entered:

- `←Login Administrator←`
- `←Login User←`

(If the password is **not** accepted, the Password prompt reappears.)

The Login Administrator response represents administrator level access, while the Login User response represents user level access. If the passwords are the same for both administrator and user, the unit defaults to administrator privileges.

If the unit is **not** password-protected, it is ready to accept SIS commands after it sends the copyright message.

Scaler-initiated Messages

When certain local events occur, such as a change in the status of a contact closure port, the scaler responds by sending a message to the host. No response is required from the host. The following messages may be sent:

- `Reconfig←` A change in the current input frequency was detected.
- `Hp1g0[X2]*[X70]←` A hot plug event was detected on output [X2].
For [X70]: 1 = assertion, 2 = de-assertion
- `HdcpI [X1]*[X44]←` A change was detected in the HDCP status of input [X1].
- `HdcpO [X2]*[X44]←` A change was detected in the HDCP status of output [X2].
- `IN00•[X61]*[X61]*[X61]*[X61]←` This message appears when a sync change (detected or removed) occurs on any input. [X61] is the video signal status for each input.
- `Cntc[X75]*[X76]` Contact closure port [X75] (1 through 7) has changed status to [X76].

See [Symbol Definitions](#) starting on page 37 for an explanation of the X variables in these messages.

SIS Commands

The DSC can be configured and controlled via a host computer or other device (such as a control system) attached to the front panel USB port or the rear panel LAN connector. Enter SIS commands using an SSH client.

SIS commands consist of one or more characters per field. No special characters are required to begin or end a command sequence. When the DSC determines that a command is valid, it executes the command and sends a response to the host device. All responses from the scaler to the host end with a carriage return and a line feed (CR/LF = `←`), indicating the end of the response character string (one or more characters).

DSC-initiated (Unsolicited) Messages

When a local event such as a front panel selection or adjustment takes place, the DSC scaler responds by sending a message to the host. No response is required from the host. DSC scaler-initiated messages include:

- `(c) Copyright 20nn, Extron, {DSC 12G-HD A or DSC HD-12G A}, Vn.nn, 60-206n-01←`
`{Day, DD MM YYYY, hh:mm:ss}←`

This message displays when the unit is first powered up on a newly connected internet protocol (IP) port.

- 20nn is the release year of the firmware.
- Vn.nn is the firmware version number.
- 60-206n-01 is the unit part number:
60-2060-01 = DSC HD-12G A, 60-2061-01 = DSC 12G-HD A.

- Reconfig ←
This message is broadcast when a change in the input frequency is detected.
- Hplg01*1 and Hdcp01*1 (DSC 12G-HD A only)
These messages are broadcast when a HDCP sink is connected or powered on.
- Hplg01*2 and Hdcp01*0 (DSC 12G-HD A only)
These messages are broadcast when an HDMI sink is disconnected or powered off.

Error Responses

When the DSC receives a valid command, it executes the command and sends a response to the host device. If the unit is unable to execute the command because the command contains invalid parameters, it returns one of the following error responses to the host.

E01 — Invalid input number
 E10 — Invalid command
 E11 — Invalid preset number
 E12 — Invalid port number
 E13 — Invalid parameter
 E14 — Invalid command for this configuration
 E17 — Invalid command for signal type
 E22 — Busy
 E28 — Bad filename or file not found
 E33 — Bad file type or size

Using the Command and Response Table

The [Command and Response Table for SIS Commands](#) starting on page 43 lists the commands that the DSC scaler recognizes as valid, the responses that are returned to the host, a description of the command function or the results of executing the command, and an example of each command in ASCII.

NOTE: Unless otherwise stated, upper- and lowercase text can be used interchangeably in the commands.

ASCII to HEX Conversion Table										Esc 1B	CR 0D	LF 0A
Space 20	!	21	"	22	#	23	\$	24	%	25	&	26
(28	)	29	*	2A	+	2B	,	2C	-	2D	.	2E
0 30	1	31	2	32	3	33	4	34	5	35	6	36
8 38	9	39	:	3A	;	3B	<	3C	=	3D	>	3E
@ 40	A	41	B	42	C	43	D	44	E	45	F	46
H 48	I	49	J	4A	K	4B	L	4C	M	4D	N	4E
P 50	Q	51	R	52	S	53	T	54	U	55	V	56
X 58	Y	59	Z	5A	[	5B	\	5C	]	5D	^	5E
` 60	a	61	b	62	c	63	d	64	e	65	f	66
h 68	i	69	j	6A	k	6B	l	6C	m	6D	n	6E
p 70	q	71	r	72	s	73	t	74	u	75	v	76
x 78	y	79	z	7A	{	7B		7C	}	7D	~	7E
											DEL	7F

Figure 25. ASCII to Hexadecimal Character Conversion Table

Symbol Definitions

• = Space

↵ = Carriage return with line feed

← or | = Carriage return with no line feed

[Esc] or W = Escape

²⁴ = Superscripts indicate the error message displayed if the command is entered incorrectly or with invalid parameters (see Error Responses).

NOTE: Values with an asterisk (*) depend on the resolution.

[X1] = Input number

DSC 12G-HD A:

1 = SDI

DSC HD-12G A:

1 = HDMI

[X2] = Output number

DSC 12G-HD A:

1 = HDMI

DSC HD-12G A:

1 = SDI A

2 = SDI B

[X3] = Input video format

DSC 12G-HD A:

0 = No signal detected on the input

2 = HD-SDI (720p, 1080i, 1080p up to 30 Hz)

3 = 3G-SDI (1080p @ 50 Hz-60 Hz, 2048x1080p @ 50 Hz-60 Hz)

4 = 6G-SDI (4K up to 30 Hz)

5 = 12G-SDI (4K @ 50 Hz-60 Hz)

DSC HD-12G A:

0 = No signal detected on the input

1 = HDMI

2 = DVI

3 = Invalid signal

[X6] = Total lines

Response is four digits, padded with zeros.

[X7] = Total pixels

Response is four digits, padded with zeros.

[X8] = Active pixels

Response is four digits, padded with zeros.

[X9] = Active lines

Response is four digits, padded with zeros.

[X10] = On or off (enable or disable)

0 = Off or disable

1 = On or enable

[X13] = Horizontal and vertical frequencies

Format is three digits with a single decimal place and leading zeros. Example: 075.3

[X14] = Text label or preset name (up to 32 characters):

Default input preset name = Input Preset *nnn* (example: Input Preset 005)

Default logo name: Logo *XX* (example: Logo Preset 05)

NOTE: Input presets saved without a name are saved with the default name.

[X15] = Picture adjustments — 0 to 127 (default 64). Response is three digits, padded with zeros.

[X16] = Horizontal and vertical position

Depends on resolutions (±4096 horizontally, ±2160 vertically for logos)

Response is five digits with leading + or – and padded with zeros.

X17 = Horizontal and vertical size (depends on resolutions)
Response is five digits with leading + or – and padded with zeros.
Response five digits, padded with zeros.

X21 = **DSC 12G-HD A:** Scaler resolution
DSC HD-12G A: EDID/SDI output resolution
Response is three digits, padded with zeros.

X21 SIS Variables for DSC 12G-HD A Resolutions and Refresh Rates								
Resolution	23.98 Hz	24 Hz	25 Hz	29.97 Hz	30 Hz	50 Hz	59.94 Hz	60 Hz
640x480								10
800x600								11
1024x768								12
1280x768								13
1280x800								14
1280x1024								15
1360x768								16
1366x768								17
1440x900								18
1400x1050								19
1600x900								20
1680x1050								21
1600x1200								22
1920x1200								23
480p							24	25
576p						26		
720p			29	30	31	32	33	34
1080i						35	36	37
1080p	38	39	40	41	42	43	44	45*
2048x1080 (2K)	46	47	48	49	50	51	52	53
2048x1200								54
2048x1536								55
2560x1080								56
2560x1440								57
1560x1600								58
3840x2160	59	60	61	62	63	64	65	66
4096x2160	69	70	71	72	73	74	75	76
Custom Output rates				201 - 205				

*Default output resolution

X21 SIS Variables for DSC HD-12G A EDID / Output Resolution and Refresh Rates								
Automatic — Match scaler current output resolution***					0			
Resolution	23.98 Hz	24 Hz	25 Hz	29.97 Hz	30 Hz	50 Hz	59.94 Hz	60 Hz
640x480								10
800x600								11
1024x768								12
1280x768								13
1280x800								14
1280x1024								15
1360x768								16
1366x768								17
1440x900								18
1400x1050								19
1600x900								20
1680x1050								21
1600x1200								22
1920x1200								23
480p							24	25
576p						26		
720p			29	30	31	32	33	34
1080i						35	36	37
1080p	38	39	40	41	42	43	44	45*
2048x1080 (2K)	46	47	48	49	50	51	52	53
2048x1200								54
2048x1536								55
2560x1080								56
2560x1440								57
1560x1600								58
3840x2160	59	60	61	62	63	64***	65***	66***
4096x2160*****	69	70	71	72	73	74	75	76
Custom Output rates				201 - 205				

*Default output resolution

***Default input EDIDs

*****4096x2160 rates are available only for output resolution and cannot be selected for input EDIDs.

NOTE: Only the rates in the blue cells are available as scaled SDI output rates.

- X22** = Test patterns:
0 = Off (default)
1 = Crop
2 = Alternating pixels
3 = Crosshatch
4 = Color bars
5 = Grayscale
6 = Audio Test (Crop pattern, OSD text "AUDIO TEST." Outputs pink noise at 48 kHz, 24 bits, channels 1/2.)
Response is two digits, padded with a zero.
- X26** = Input presets — 1 to 128 (global for device).
Response is three digits padded with a zero.

- X28** = Output sync timeout — 1 to 500 in 1-second increments
 0 = The output sync is instantly disabled with no active input.
 501 = The output sync never times out (default).
- X29** = Front panel lock (executive mode) status
 0 = Disabled (default)
 1 = Enabled (front panel locked)
- X30** = OSD menu system status
 0 = Off
 1 = On
- X31** = Verbose mode
 0 = None (default for USB and LAN connections).
 1 = Verbose mode is on.
 2 = Verbose mode is off, tagged responses are sent for queries (tagged responses are enabled).
 3 = Verbose mode is on (enabled) and tagged responses are enabled and sent for queries.

NOTES:

- In verbose response mode, the DSC responds with more information than it usually would. For example, the DSC can send out a notice of a change in some setting without receiving a query via your PC. That change could have been a result of an internal process (a script execution), a selection made using a touchpanel or keypad, a change made using GV or a web page, or input from a connected sensor or switch.
 That is an example of a verbose (wordy) relationship between the DSC and a connected device. Verbose mode creates more network traffic than usual, which can slow down network performance.
- If tagged responses are enabled, all read or view commands return the constant string plus the data.
Example:
Command: `[Esc] M1SSAV` ←
Response: `SsavM1*[X40]` ← (tagged mode) or `[X40]` ← (untagged mode)

- X32** = Unit name — A text string of up to 63 characters (A-Z, a-z, 0-9, and the hyphen (-) are permitted).
 - The first character must be a letter.
 - The last character **cannot** be a hyphen.
 - No blank or space characters are permitted as part of the name.
 - No distinction is made between uppercase and lowercase letters.
- X33** = Default name — DSC 12G-HD-A-XX-XX-XX **or** DSC HD-12G-A-XX-XX-XX
 XX-XX-XX is the last three character pairs of the device MAC address.
- X39** = Aspect ratio
 1 = Fill — Each input rate automatically fills the entire output raster (default)
 2 = Follow — Each input rate is displayed with its native aspect ratio
- X40** = Screen saver mode
 1 = Black screen (default)
 2 = Blue output with OSD text
 3 = User image on a black screen
- X42** = Video mute
 0 = Off (unmute) (default)
 1 = On (mute to black screen)
 2 = On (mute all output sync and video)
- X43** = User logo 001 through 016.
 101 = No signal, screen saver user logo, or image
 Response is three digits.
- X45** = Upstream video switching effect
 0 = Cut through black
 1 = Fade through black
 2 = Seamless fade
 3 = Seamless cut
 4 = Low latency: Removes additional scaler frame buffer (default).

- X48** = Output format
- DSC 12G-HD A: HDMI output format**
- 0 = Auto (default). (HDMI – RGB Full to a CEA sink or DVI to a non-CEA sink)
 - 1 = DVI (RGB 444, 0-255, no audio, no InfoFrames) (165 MHz maximum output)*
 - 2 = HDMI RGB Full (RGB 444, 0-255, audio, InfoFrames)
 - 3 = HDMI RGB Limited (RGB 444, 16-235, audio, InfoFrames)
 - 5 = HDMI YUV Limited (YUV 444, 16-235, audio, InfoFrames)
 - 7 = HDMI YUV Limited (YUV 422, 16-235, audio, InfoFrames)
- *If in DVI mode and a rate greater than 165 MHz is selected, the output defaults to HDMI RGB 444 Full.
- DSC HD-12G A: SDI output format**
- 2 = HD-SDI
 - 3 = 3G-SDI
 - 4 = 6G-SDI
 - 5 = 12G-SDI
- X50** = Output volume
- 0 dB to -100 dB in 1.0 dB steps
 - 0 dB = default.
- Response is three digits with a leading + or - sign. (*Example: +000 or -010*)
- X51** = Audio output format
- 1 = Dual mono
 - 2 = Stereo (default)
- X52** = SDI audio channel, pair 1 or 2
- Default = 1
- Response is one digit.
- X53** = SDI audio group — AES channel pair group 1 or 2
- 1 = Group 1
 - 2 = Group 2
- X54** = Audio input gain
- 18 through +24 dB in 1.0 dB steps. Default = 0 dB.
- Response is two digits with leading + or - and is padded with a zero.
- X55** = Swap channel
- 0 = Off
 - 1 = On
- X58** = Audio input format
- 0 = None (input audio muted)
 - 1 = Analog (on 5-pole captive screw connector)
 - 2 = LPCM-2Ch (default)
 - 3 = Multi-Ch
- X60** = EDID file data block, 128 or 256 bytes of binary data (machine-readable only).
- X61** = Video signal status
- 0 = Video signal not detected
 - 1 = Video signal detected.
- X62** = Power save mode
- 0 = Full power mode (default)
 - 1 = Low power mode
- X63** = Screen saver status
- 0 = Active input detected, timer not running
 - 1 = No active input detected, timer running, output sync active
 - 2 = No active input detected, timer expired, output sync disabled
- X64** = Input lock (AFL) setting
- DSC 12G-HD A**
- 0 = Disable input lock (default). Free running pixel clock is generated internally.
 - 1 = Enable input lock (AFL). Lock output vertical to vertical refresh rate of the input.
- DSC HD-12G A**
- 0 = Disable input lock (default). Free running pixel clock is generated internally.
 - 1 = Enable input lock (AFL). Lock output vertical to vertical refresh rate of the input.
 - 2 = Enable SDI Genlock. Lock the output vertical to the applied Genlock input.

- X65 = Input lock (AFL) status — DSC 12G-HD A**
 0 = Input lock (AFL) is disabled.
 1 = Input lock (AFL) is enabled, but cannot lock to the applied input signal. The DSC defaults to the set output rate with a free-running clock.
 2 = Input lock (AFL) is enabled. The output is locked to the applied input signal (AFL) or SDI input.
- Input lock (AFL) / SDI Genlock status — DSC HD-12G A**
 0 = Input lock (AFL) / SDI Genlock disabled (default).
 1 = Input lock (AFL) / SDI Genlock is enabled, but cannot lock to the applied input signal (AFL) / SDI Genlock input. The DSC defaults to the set output rate..
 2 = Input lock (AFL) / SDI Genlock is enabled. The output is locked to the applied input signal (AFL) / SDI Genlock input.
- X66 = Genlock pixel offset — DSC HD-12G A only**
 Range is $\pm$ the current output resolution total pixel value -1. (*Example:* -2199 to +2199 for 1080p)
- X67 = Genlock line delay — DSC HD-12G A only**
 Range is $\pm$ the current output resolution total line value -1. (*Example:* -1124 to +1124 for 1080p)
- X69 = Hot plug change or detect**
 1 = A new source has been connected (assertion).
 2 = A source has been disconnected (de-assert).
- X70 = Key effect variable**
 0 = Transparency (available only when Key effect **X74** = 1)
 1 = Red of RGB key (available only when Key effect **X74** = 2)
 2 = Green of RGB key (available only when Key effect **X74** = 2)
 3 = Blue of RGB key (available only when Key effect **X74** = 2)
 4 = Level Key (available only when Key effect **X74** = 1)
- X71 = Key effect level setting (0 - 255)**
- X72 = Audio output**
 1 = Local 5-pole captive screw analog audio output
 2 = Embedded HDMI or SDI audio output
- X74 = Key effect**
 0 = Disabled
 1 = Transparency
 2 = RGB key
 3 = Level key
 4 = Alpha key
- X83 = SIS over SSH echo status**
 0 = Disabled
 1 = Enabled (default)
 Each character that is transmitted out is sent back in. Extra CR (↵) and CR/LF (↵) characters may appear as well.
- X88 = Current internal temperature**
- X89 = Over-temperature delta**
 Number of degrees Celsius that the internal temperature is above or below the product-specific threshold, to one decimal place (preceded by a + or - sign)
- X91 = Administrator or user password**
 • Length is 1-128 characters.
 • All human-readable characters are allowed except |.
 • The password cannot be a single space.
 • Passwords are case-sensitive.
 • A user password cannot be assigned if no administrator password exists. An E14 error code is returned.
 • If the admin password is cleared, the user password is cleared as well.
- X97 = Upstream switch masking duration (in milliseconds)**
 100 - 10000 in 1 ms steps (default = 200 ms)

Command and Response Table for SIS Commands

Command	ASCII Command (Host to Scaler)	Response (Scaler to Host)	Additional Description
Input Configuration			
Input Video Format			
View video format	1*\	X3 ↵ <i>In verbose modes 2 and 3:</i> Vtyp1* X3 ↵	View detected input video format X3 .
KEY: X3 = Input video format for DSC 12G-HD A Input video format for DSC HD-12G A		0 = No signal, 2 = HD-SDI, 3 = 3G-SDI, 4 = 6G-SDI, 5 = 12G-SDI 0 = No signal, 1 = HDMI, 2 = DVI	
Input EDID (DSC HD-12G A only)			
NOTE: {Filename} can be a full path name with a .bin extension, carrying 128 or 256 bytes of binary data.			
Assign an EDID to an input	[Esc] A 1* X21 EDID ←	Edid A 1* X21 ↵	Assign EDID resolution and refresh rate X21 to the DSC HD-12G A input.
View assigned EDID	[Esc] A 1 EDID ←	X21 ↵ <i>In verbose modes 2 and 3:</i> Edid A 1 * X21 ↵	View assigned EDID resolution and refresh rate X21 for the DSC HD-12G A input.
Export an EDID file	[Esc] E X21 , {filename}EDID ←	EdidE X21 ↵	Export EDID X21 to {filename}.
NOTE: Exporting a default EDID table (X21 value of 10 or greater) results in the digital 2-Ch audio table being exported.			
Import an EDID file	[Esc] I X21 , {filename}EDID ←	Edid I X21 ↵	Import EDID X21 from {filename}.
KEY: X21 = EDID emulations for DSC HD-12G A:		0 = Automatic: matches current output resolution (default), 1 = Custom EDID 1, 2 = Custom EDID 2, 3 = Custom EDID 3, 4 = Custom EDID 4, 5 = Custom EDID 5 For 10-76, see Scaler resolution tables starting on page 38.	
Save Output EDID (DSC 12G-HD A only)			
NOTE: {Filename} can be a full path name with a .bin extension, carrying 128 or 256 bytes of binary data.			
Save an output EDID	[Esc] S 1* X21 EDID ←	Edid S 1* X21 ↵	Save output 1 EDID to X21 . Valid only when X21 = 201-205 (custom EDIDs).
Export an EDID file	[Esc] E X21 , {filename}EDID ←	EdidE X21 ↵	Export EDID X21 to {filename}.
NOTE: Exporting a default EDID table (X21 value of 10 or greater) results in the digital 2-Ch audio table being exported.			
Import an EDID file	[Esc] I X21 , {filename}EDID ←	Edid I X21 ↵	Import EDID X21 from {filename}.
KEY: X21 = Custom EDIDs for DSC HD-12G A		0 = Automatic: matches current output resolution (default), 1 = Custom EDID 201, 2 = Custom EDID 220, 3 = Custom EDID 203, 4 = Custom EDID 204, 5 = Custom EDID 205 For 10-76, see SIS Variables for DSC HD-12G A EDID on page 39.	

Command	ASCII Command (Host to Scaler)	Response (Scaler to Host)	Additional Description
Input Configuration (continued)			
Image Reset			
Execute	1*0A	Img1*0↵	Execute an Image Reset on the current input (follows current aspect ratio setting).
Execute and fill	1*1A	Img1*1↵	Execute an Image Reset and fill entire display.
Execute and follow	1*2A	Img1*2↵	Perform an Image Reset and follow the aspect ratio of the input.
Active Pixels			
View	[Esc] 1 APIX ←	[X8]↵ In verbose modes 2 and 3: Apix1*[X8]↵	Show active pixels [X8] for the input.
Active Lines			
View	[Esc] 1 ALIN ←	[X9]↵ In verbose modes 2 and 3: Alin1*[X9]↵	Show active lines [X9] for the input. [X9] is four digits padded with zeros.
Total Lines			
View total lines	[Esc] 1 TLIN ←	[X6]↵ In verbose modes 2 and 3: Tlin 1*[X6]↵	Show total number of lines for the input.
Total Pixels			
View total pixels	[Esc] 1 TPIX ←	[X7]↵ In verbose modes 2 and 3: Tpix 1*[X7]↵	Show the total number of pixels ([X7]) for the input.
KEY: [X8] = Active pixels [X6] = Total lines [X7] = Total pixels [X9] = Active lines		Response is four digits padded with zeros. Response is four digits padded with zeros. Response is four digits padded with zeros. Response is four digits padded with zeros..	
3:2, 2:2, and 24:1 Film Mode Detection			
Enable auto detection	[Esc]1*1 FILM ←	Film 1*1↵	Enable automatic detection of film mode (3:2, 2:2, and 24:1 pull-down) (default).
Disable auto detection	[Esc]1*0 FILM ←	Film 1*0↵	Disable automatic film mode detection.
View setting	[Esc]1FILM ←	[X10]↵ In verbose modes 2 and 3: Film 1*[X10]↵	View the current film mode setting.
KEY: [X10] = Film mode automatic detection status		0 = Disabled, 1 = Enabled	

Command	ASCII Command (Host to Scaler)	Response (Scaler to Host)	Additional Description
Input Configuration (continued)			
Switch Effect — Upstream			
Set switch effect	[Esc] U 1*[X45]SWEF ←	Swef U 1*[X45]↵	Enable upstream switch effect [X45].
View switch effect	[Esc] U 1 SWEF ←	[X45]↵ <i>In verbose modes 2 and 3:</i> Swef U 1*[X45]↵	View upstream switch effect setting.
Upstream Switch Masking			
Set upstream switch masking	[Esc] H 1*[X97]SWEF ←	Swef H 1*[X97]↵	Set upstream switch masking time to [X97] milliseconds.
View upstream switch masking	[Esc] H 1 SWEF ←	[X97]↵ <i>In verbose modes 2 and 3:</i> Swef H 1*[X97]↵	View upstream switch masking setting.
KEY: [X45] = Upstream switch effect [X97] = Upstream switch masking duration 0 = Cut through black, 1 = Fade through black, 2 = Seamless fade, 3 = Seamless cut, 4 = Low latency: Removes additional scaler frame buffer (default). 100-10000 milliseconds in 1 ms steps. Default = 200 ms			
Picture Adjustments			
Video Mute (Reverts to Unmuted after a power cycle.)			
Mute all video to black	1B	Vmt1↵	Mute video to black on all outputs.
Mute all sync and video	2B	Vmt2↵	Mute video and sync on all outputs.
Unmute all video and sync	0B	Vmt0↵	Unmute and display the video on all outputs.
View	B	[X42]↵ <i>In verbose modes 2 and 3:</i> Vmt[X42]↵	View output mute status [X42].
KEY: [X42] = Video mute status 0 = Unmuted (default), 1 = Muted, 2 = Output sync and video muted			
Contrast			
Set a specific value	[Esc] 1*[X15] CONT ←	Cont 1*[X15]↵	Set the range of image light and dark values (contrast level) to [X15] for the input.
Increment contrast value	[Esc] 1*+CONT ←	Cont 1*[X15]↵	Increase the contrast level by 1.
Decrement value	[Esc] 1-CONT ←	Cont 1*[X15]↵	Decrease the contrast level by 1.
View	[Esc] 1CONT ←	[X15]↵ <i>In verbose modes 2 and 3:</i> Cont 1*[X15]↵	View the current contrast level for the input.
Brightness			
Set a specific value	[Esc] 1*[X15] BRIT ←	Brit 1*[X15]↵	Set the intensity of video light on the screen (brightness level) to [X15] for the input.
Increment value	[Esc] 1+BRIT ←	Brit 1*[X15]↵	Increase the brightness level by 1.
Decrement value	[Esc] 1-BRIT ←	Brit 1*[X15]↵	Decrease the brightness level by 1.
View	[Esc] 1BRIT ←	[X15]↵ <i>In verbose modes 2 and 3:</i> Brit 1*[X15]↵	View the current brightness level for the input.
KEY: [X15] = Picture adjustments (contrast or brightness) 0 to 127 (default = 64). Response is five digits with leading + or - and padded with zeros.			

Command	ASCII Command (Host to Scaler)	Response (Scaler to Host)	Additional Description
Picture Adjustments (continued)			
Detail Filter			
Set detail level	[Esc] 1* [X15] HDET ←	Hdet 1* [X15] ↵	Set the sharpness (detail) level to [X15] for the input.
Increment value	[Esc] 1 +HDET ←	Hdet 1* [X15] ↵	Increase the detail level by 1.
Decrement value	[Esc] 1 -HDET ←	Hdet 1* [X15] ↵	Decrease the detail level by 1.
View detail value	[Esc] HDET ←	[X15] ↵ <i>In verbose modes 2 and 3:</i> Hdet 1* [X15] ↵	Show the detail level for the input.
Horizontal Shift — Image			
Set horizontal position to a specific value	[Esc] I1* [X16] HCTR ←	Hctr I1* [X16] ↵	Set the horizontal position of the image in relation to the center of the screen to [X16] .
Increment value	[Esc] I1 +HCTR ←	Hctr I1* [X16] ↵	Shift image right one pixel.
Decrement value	[Esc] I1 -HCTR ←	Hctr I1* [X16] ↵	Shift image left one pixel.
View	[Esc] I1 HCTR ←	[X16] ↵ <i>In verbose modes 2 and 3:</i> Hctr I1* [X16] ↵	View image horizontal position [X16] .
Vertical Shift — Image			
Set vertical position to a specific value	[Esc] I1* [X16] VCTR ←	Vctr I1* [X16] ↵	Set the vertical position of the image in relation to the center of the screen to [X16] .
Increment value	[Esc] I1 +VCTR ←	Vctr I1* [X16] ↵	Shift the image down by one line.
Decrement value	[Esc] I1 -VCTR ←	Vctr I1* [X16] ↵	Shift the image up by one line.
View	[Esc] I1 VCTR ←	[X16] ↵	View image vertical position [X16] .
Horizontal Size — Image			
Specific value	[Esc] I1* [X17] HSIZ ←	Hsiz I1* [X17] ↵	Set the horizontal size (width) of the image to [X17] .
Increase horizontal size	[Esc] I1 +HSIZ ←	Hsiz I1* [X17] ↵	Widen the image by one pixel.
Decrease horizontal size	[Esc] I1 -HSIZ ←	Hsiz I1* [X17] ↵	Make the image narrower by one pixel.
View	[Esc] I1 HSIZ ←	[X17] ↵ <i>In verbose modes 2 and 3:</i> Hsiz I1* [X17] ↵	View image horizontal size [X17] .
Vertical Size — Image			
Specific value	[Esc] I1* [X17] VSIZ ←	Vsiz I1* [X17] ↵	Set the vertical size (height) of the image to [X17] .
Increase vertical size	[Esc] I1 +VSIZ ←	Vsiz I1* [X17] ↵	Make the image taller by one line.
Decrease vertical size	[Esc] I1 -VSIZ ←	Vsiz I1* [X17] ↵	Make the image shorter by one line.
View	[Esc] I1 VSIZ ←	[X17] ↵ <i>In verbose modes 2 and 3:</i> Vsz I1* [X17] ↵	View image vertical size [X17] .
KEY: [X15] = Picture adjustment (detail filter) [X16] = Horizontal or vertical position [X17] = Horizontal or vertical size			
		0 through 127 (default = 64) Depends on resolutions (±4096 horizontally, ±2160 vertically for logos) Response is five digits with leading + or - and padded with zeros. Depends on resolutions. Response is five digits with leading + or - and padded with zeros.	

Command	ASCII Command (Host to Scaler)	Response (Scaler to Host)	Additional Description						
Picture Adjustments (continued)									
Compound Image Position and Size									
Specific value	[Esc]1, [X16]*[X16]*[X17]*[X17] XIMG←	Ximg1, [X16]*[X16]*[X17]*[X17]←	Set the x (horizontal) and y (vertical) position ([X16]), and the horizontal and vertical size ([X17]) for the image.						
View	[Esc]1 XIMG←	[X16]*[X16]*[X17]*[X17]← In verbose modes 2 and 3: Ximg1, [X16]*[X16]*[X17]*[X17]←	View x, y position and the size for image.						
KEY: [X15] = Picture adjustment (detail filter) [X16] = Horizontal or vertical position [X17] = Horizontal or vertical size 0 through 127 (default = 64) Depends on resolutions (±4096 horizontally, ±2160 vertically for logos) Response is five digits with leading + or - and padded with zeros. Depends on resolutions. Response is five digits with leading + or - and padded with zeros.									
Presets									
NOTE: Unnamed presets are shown as [unassigned]. To restore a default user preset name, enter a single space character for [X14]. These entries are valid only for previously saved presets.									
<table><tr><th>Settings Saved in Input Presets</th></tr><tr><td>Preset name</td></tr><tr><td>Horizontal size</td></tr><tr><td>Vertical size</td></tr><tr><td>Horizontal image position</td></tr><tr><td>Vertical image position</td></tr></table>				Settings Saved in Input Presets	Preset name	Horizontal size	Vertical size	Horizontal image position	Vertical image position
Settings Saved in Input Presets									
Preset name									
Horizontal size									
Vertical size									
Horizontal image position									
Vertical image position									
Auto Memories									
Enable	[Esc]1*1 AMEM←	Amem 1*1←	Set Auto Memory to On (default). Settings are automatically stored as presets. Previous settings for the incoming signal are recalled.						
Disable	[Esc]1*0 AMEM←	Amem 1*0←	Set Auto Memory to Off. Manual recall of input presets is required to configure the input.						
View setting	[Esc]1 AMEM←	[X10]← In verbose modes 2 and 3: Amem 1*[X10]←	View current Auto Memory status [X10].						
Input Presets									
Recall preset	2*[X26].	2Rpr1*[X26]←	Recall input preset [X26] for the input.						
Save preset	2*[X26],	2Spr1*[X26]←	Save input preset [X26] for the input.						
Delete preset	[Esc] X 2*[X26] PRST ←	PrstX 2*[X26]←	Clear input preset [X26] and set its name to [unassigned].						
KEY: [X10] = Enable or disable Auto Memory [X14] = Preset name [X26] = Input preset number 0 = Off or disable, 1 = On or enable (default) Up to 32 characters Default = Input Preset nnn (Example: Input Preset 005) 1 - 128 Response is three digits padded with zeros.									

Command	ASCII Command (Host to Scaler)	Response (Scaler to Host)	Additional Description
Presets (continued)			
NOTE: Unnamed presets are shown as [unassigned]. To restore a default user preset name, enter a single space character for X14. These entries are valid only for previously saved presets.			
Input Preset Name			
Write preset name	[Esc] 2* X26, X14 PNAME←	Pnam 2 *X26, X14↵	Set the name of input preset X26 to X14.
Read preset name	[Esc] 2*X26 PNAME←	X14↵ In verbose modes 2 and 3: Pnam 2*X26, X14↵	View the name of input preset X26.
KEY: X14 = Preset name X26 = Input preset number		Up to 32 characters Default = Input Preset nnn (Example: Input Preset 005) 1 - 128 Response is three digits padded with zeros.	
Output Configuration			
Output Scaler Rate			
Set output rate	[Esc] 1*X21 RATE←	Rate 1*X21↵	Select output resolution and refresh rate X21.
View output rate	[Esc] 1RATE←	X21↵ In verbose modes 2 and 3: Rate 1*X21↵	Show selected output rate.
KEY: X21 = Output resolution and refresh rate		0 = Automatic: Match scaler current output resolution (default) 1 = Matches the connected HDMI sink EDID (EDID save and export only) 10 – 76, see the Scaler resolution tables starting on page 38. Response is two digits padded with a zero. 201 - 205 = Custom EDIDs or output rates	
Power Save Mode			
Power save off	[Esc] 0 PSAV←	Psav 0↵	Set the DSC to run in full power mode (default).
Power save on	[Esc] 1 PSAV←	Psav 1↵	Place the DSC in low power mode.
View setting	[Esc] PSAV←	X62↵ In verbose modes 2 and 3: Psav X62↵	View current power save status X62.
KEY: X62 = Power save setting		0 = Full power mode (default) 1 = Low power mode	

Command	ASCII Command (Host to Scaler)	Response (Scaler to Host)	Additional Description
Output Configuration (continued)			
Screen Saver Mode (activated when no signal is detected on the video input)			
NOTE: When the screen saver mode automatically enables or disables output sync, an unsolicited SsavS X63 response is broadcast.			
Set screen saver mode	Esc M 1 * X40 SSAV ←	Ssav M 1 * X40 ↵	Sets the screen saver mode to X40 . Default = 1 (black screen).
View screen saver mode	Esc M 1 SSAV ←	X40 ↵ <i>In verbose modes 2 and 3:</i> SsavM1 * X40 ↵	View current screen saver mode X40 .
Set timeout duration	Esc T 1 * X28 SSAV ←	SsavT1 * X28 ↵	Set the period before timeout to X28 seconds. Default = 501 seconds (never).
View timeout duration	Esc T 1 SSAV	X28 ↵ <i>In verbose modes 2 and 3:</i> SsavT 1 * X28 ↵	View amount of time X28 before sync timeout.
View screen saver status	Esc S 1 SSAV ←	X63 ↵ <i>In verbose modes 2 and 3:</i> SsavS 1 * X63 ↵	View the screen saver status X63 .
KEY: X28 = Output sync timeout — 1 through 500, in 1-second increments (default = 501 seconds = never) 0 = The output sync is instantly disabled with no active input. 501 = Output sync never times out (default). X40 = Screen saver mode: 1 = black screen (default), 2 = blue output with OSD text X63 = Screen saver status: 0 = Active input detected, timer not running 1 = No active input detected, timer running, output sync active 2 = No active input detected, timer expired, output sync disabled			
Output Format			
Set HDMI output format	Esc 1 * X48 VTPO ←	Vtpo X48 ↵	Set output format for the output to X48 .
View HDMI or SDI output format	Esc 1VTPO ←	X48 ↵	Show current HDMI or SDI output format setting.
KEY: X48 = HDMI output format — DSC 12G-HD A 0 = Auto (default). (HDMI – RGB Full to a CEA sink or DVI to a non-CEA sink). 1 = DVI (RGB 444, 0-255, no audio, no InfoFrames).* 2 = HDMI RGB Full (RGB 444, 0-255, audio, InfoFrames.) 3 = HDMI RGB Limited (RGB 444, 16-235, audio, InfoFrames.) 5 = HDMI YUV Limited (YUV 444, 16-235, audio, InfoFrames.) 7 = HDMI YUV Limited (YUV 422, 16-235, audio, InfoFrames.) SDI output format — DSC HD-12G A 2 = HD-SDI, 3 = 3G-SDI, 4 = 6G-SDI, 5 = 12G-SDI			

Command	ASCII Command (Host to Scaler)	Response (Scaler to Host)	Additional Description
Output Configuration (continued)			
Output Status (DSC 12G-HD A only)			
Query output	[Esc] 01HDCP ←	[X61] ←	View whether a sink device is detected (HDCP status).
AFL (Accu-Frame Lock) Input Lock or SDI Genlock			
Disable AFL	[Esc] 1*0 GLOK ←	Glok 1*0 ←	Disable global AFL input lock (default).
Enable AFL (input lock)	[Esc] 1*1 GLOK ←	Glok 1*1 ←	Lock the output refresh rate to the rate of the selected input.
Enable SDI genlock (DSC HD-12G A only)	[Esc] 1*2 GLOK ←	Glok 1*2 ←	Lock the output refresh rate to the SDI genlock rate.
View the AFL or genlock setting	[Esc] 1 GLOK ←	[X64] ← <i>In verbose modes 2 and 3:</i> Glok 1*[X64] ←	View the current global input lock (AFL) or genlock setting.
View AFL or genlock status	[Esc] 41 STAT ←	[X65] ← <i>In verbose modes 2 and 3:</i> 41 Stat • [X65] ←	View the current global input lock (AFL) or genlock status.
Genlock Offset (DSC HD-12G A only)			
NOTES: - Genlock offset commands return an error code unless SDI genlock is currently enabled and genlock is currently locked to an applied genlock reference. - Genlock offsets apply only to the current output resolution and reset to 0, 0 when the output resolution is adjusted.			
Set horizontal offset	[Esc] H x GLOF ←	Glof H x ←	Set the horizontal genlock offset to x pixels. The x value (genlock pixel offset) range is ± the current output resolution total pixel value -1. <i>Example:</i> -2199 to +2199 for 1080p
View horizontal offset	[Esc] H GLOF ←	x ← <i>In verbose modes 2 and 3:</i> Glof V x ←	View the horizontal genlock offset (default = 0).
Set vertical offset	[Esc] V y GLOF ←	Glof V y ←	Set the vertical genlock offset to y pixels. The y value (genlock pixel offset) range is ± the current output resolution total line value -1. <i>Example:</i> -1124 – +1124 for 1080p
View vertical offset	[Esc] V GLOF ←	y ← <i>In verbose modes 2 and 3:</i> Glof V y ←	View the vertical genlock offset (default = 0).
KEY: [X44] = HDCP status: 0 = no sink device detected, 2 = sink detected [X64] = Input lock (AFL) status: 0 = Input lock disabled (default) 1 = Input lock is enabled but cannot lock to applied input signal (defaults to the set output rate). 2 = Input lock is enabled. The output is locked to the applied input signal (AFL). [X65] = Input lock (AFL) status — DSC 12G-HD A 0 = Input lock (AFL) is disabled. 1 = Input lock (AFL) is enabled, but cannot lock to the applied input signal. The DSC defaults to the set output rate with a free-running clock. 2 = Input lock (AFL) is enabled. The output is locked to the applied input signal (AFL) or SDI input. Input lock (AFL) / SDI Genlock status — DSC HD-12G A 0 = Input lock (AFL) / SDI Genlock disabled (default). 1 = Input lock (AFL) / SDI Genlock is enabled, but cannot lock to the applied input signal (AFL) / SDI Genlock input. The DSC defaults to the set output rate. 2 = Input lock (AFL) / SDI Genlock is enabled. The output is locked to the applied input signal (AFL) / SDI Genlock input.			

Command	ASCII Command (Host to Scaler)	Response (Scaler to Host)	Additional Description
Logos			
User Supplied Logo Image			
Select logo image file	[Esc] A [X43] , {fLenam}LOGO←	LogoA [X43] , {fLenam}←↵	Assign logo {fLenam} to logo slot [X43] .
View logo image file	[Esc] A [X43] LOGO←	{fLenam}←↵ In verbose modes 2 and 3: LogoA {fLenam} [X43] ←↵	View file name assigned to logo [X43] .
Clear Logo Setting			
Clear logo	[Esc] X3* [X43] PRST←	Prst X3* [X43] ←↵	Clear logo [X43] and set logo name to [unassigned].
Logo On and Off			
Disable logo	[Esc] E 1*0LOGO←	Logo E 1*0←↵	Disable full screen notification logo
Enable logo	[Esc] E 1* [X43] LOGO←	Logo E 1* [X43] ←↵	Enable full screen notification logo preset [X43] .
View logo status	[Esc] E 1LOGO←	[X43] ←↵ In verbose modes 2 and 3: LogoE1* [X43] ←↵	View full screen notification logo status.
Logo Name			
Write logo name	[Esc] L [X43] , [X14] UNAM←	UnamL [X43] , [X14] ←↵	Set name for logo [X43] .
View logo name	[Esc] L [X43] UNAM←	[X14] ←↵ In verbose modes 2 and 3: UnamL [X43] , [X14] ←↵	View name for logo [X43] .
Logo Availability			
Query availability	[Esc] Q LOGO←	{16 characters}*{1 character}*{1 character}←↵ 0 = Empty, 1 = Saved. Example: 000...0*0*0.	
NOTE: The first 16 digits represent logo images. The digit immediately following the first * is for the screen saver logo. The last digit is for the HDCP logo.			
Horizontal Shift — Logo			
Set logo horizontal position to a specific value	[Esc] L [X43] * [X16] HCTR←	Hctr L [X43] * [X16] ←↵	Set the horizontal position of the logo in relation to the center of the screen to [X16] .
Increment value	[Esc] L [X43] +HCTR←	Hctr L [X43] * [X16] ←↵	Shift the logo right.
Decrement value	[Esc] L [X43] -HCTR←	Hctr L [X43] * [X16] ←↵	Shift the logo left.
View	[Esc] L [X43] HCTR←	[X16] ←↵ In verbose modes 2 and 3: Hctr L [X43] * [X16] ←↵	View logo horizontal position [X16] .
Vertical Shift — Logo			
Set logo vertical position to a specific value	[Esc] L [X43] * [X16] VCTR←	Vctr L [X43] * [X16] ←↵	Set the vertical position of the logo in relation to the center of the screen to [X16] .
Increment value	[Esc] L [X43] +VCTR←	Vctr L [X43] * [X16] ←↵	Shift the logo down.
Decrement value	[Esc] L [X43] -VCTR←	Vctr L [X43] * [X16] ←↵	Shift the logo up.
View	[Esc] L [X43] VCTR←	[X16] ←↵ In verbose modes 2 and 3: Vctr L [X43] * [X16] ←↵	View logo vertical position [X16] .
KEY: [X14] = Logo name [X16] = Horizontal or vertical position [X43] = User logo			
Up to 32 characters. Default = Logo XX. (Example: Logo Preset 05.) Depends on resolutions (±4096 horizontally, ±2160 vertically for logos) Response is five digits with leading + or - and padded with zeros. 001 through 016 101 = No signal, screen saver, user logo, or image			

Command	ASCII Command (Host to Scaler)	Response (Scaler to Host)	Additional Description
Logos (continued)			
Logo Key Effect			
Disabled	[Esc] [X43]*0 LKEF←	Lkef [X43]*0 ←	Disable key effect for logo [X43]
Transparency	[Esc] [X43]*1 LKEF←	Lkef [X43]*1 ←	Enable transparency for logo [X43] .
RGB key	[Esc] [X43]*2 LKEF←	Lkef [X43]*2 ←	Enable RGB key for logo [X43] .
Level key	[Esc] [X43]*3 LKEF←	Lkef [X43]*3 ←	Enable Level key for logo [X43] .
Alpha key	[Esc] [X43]*4 LKEF←	Lkef [X43]*4 ←	Enable Alpha key for logo [X43] .
View setting	[Esc] [X43] LKEF←	[X43] ←	View current setting [X74] for logo [X43] .
Logo Key Effect Level			
Specific level value	[Esc] [X43]*[X70]*[X71] LKEY←	Lkey [X74]*[X70]*[X71] ←	Set key level for setting [X70] to [X71] for logo [X43] .
View level setting	[Esc] [X43]*[X70] LKEY←	[X71] ←	View current [X70] setting for logo [X43] .
KEY: [X43] = User logo [X70] = Key effect variable [X71] = Key effect level setting [X74] = Key effect			
001 through 016, 101 = No signal, screen saver user logo, or image Response is three digits. 0 = Transparency (available only when Key effect [X74] = 1) 1 = Red of RGB key (available only when Key effect [X74] = 2) 2 = Green of RGB key (available only when Key effect [X74] = 2) 3 = Blue of RGB key (available only when Key effect [X74] = 2) 4 = Level Key (available only when Key effect [X74] = 1) 0 - 255 0 = Disabled, 1 = Transparency, 2 = RGB key, 3 = Level key, 4 = Alpha key			
Audio Configuration			
Audio Mute — SDI or HDMI Out and Analog Out Setting (persists through a power cycle)			
Enable global audio mute	1Z	Amt1←	Mute all audio output.
Disable global audio mute	0Z	Amt0←	Unmute all audio outputs.
Enable or disable discrete mute	[X72]*[X10]Z	Amt [X72]*[X10] ←	Mute or unmute audio output [X72] .
View individual mute status	[X72]*Z	[X10] ← <i>In verbose modes 2 and 3:</i> Amt [X10] ←	View audio mute status [X10] of audio output [X72] .
View output mute status	Z	[X10] • [X10] ← <i>In verbose modes 2 and 3:</i> Amt [X10] • [X10] ←	View output audio mute status (<i>AnaLog output • SDI output or AnaLog output • HDMI output</i>)
Audio Gain and Attenuation (analog only)			
Set audio gain and attenuation	[X54] G	Aud [X54] ←	Set the audio gain or attenuation on the analog input to [X54] dB.
Increment	+G	Aud [X54] ←	Increase the audio level by 1 dB.
Decrement	-G	Aud [X54] ←	Decrease the audio level by 1 dB.
View	G	[X54] ← <i>In verbose modes 2 and 3:</i> Aud [X54] ←	View the current input audio level.
KEY: [X10] = Mute or unmute audio [X54] = Audio gain and attenuation [X72] = Audio output			
1 = Unmute (default), 1 = Mute -18 through +12 dB Response is two digits with leading + or – and padded with a zero. 1 = Local 5-pole captive screw analog audio output 2 = Embedded SDI or HDMI audio output			

Command	ASCII Command (Host to Scaler)	Response (Scaler to Host)	Additional Description
Audio Configuration (continued)			
Audio Input Format			
Set input audio format	[Esc] I 1*[X58] AFMT ←	AfmtI 1*[X58] ←	Set audio format for the input to [X58].
View audio input format	[Esc] I 1AFMT ←	[X58] ← <i>In verbose modes 2 and 3:</i> AfmtI 1*[X58] ←	Show the format ([X58]) of the analog audio input.
Output Volume			
Set volume	[X50] V	Vol [X50] ←	Set the output volume to [X58].
Increment	+V	Aud [X50] ←	Increase the audio level by 1 dB.
Decrement	-V	Aud [X50] ←	Decrease the audio level by 1 dB.
View volume level	V	[X50] ← <i>In verbose modes 2 and 3:</i> Vol [X50] ←	View the current volume level setting.
Audio Output Format			
Set audio output format	[Esc] O 1*[X51] AFMT ←	AfmtO 1*[X51] ←	Select audio output format [X51].
View audio output format	[Esc] O 1AFMT ←	[X51] ← <i>In verbose modes 2 and 3:</i> AfmtO 1*[X51] ←	View selected audio output format [X51].
AES Audio Pair/Channel Selection (DSC 12G-HD A only)			
Select pair/channel	[Esc] 1*[X52] AESC ←	Aesc 1*[X52] ←	Select pair/channel [X52] for input 1 to be decoded.
View pair/channel	[Esc] 1 AESC ←	[X52] ← <i>In verbose modes 2 and 3:</i> Aesc 1*[X52] ←	
AES Audio Channel Group Selection (DSC 12G-HD A only)			
Select channel group	[Esc] 1*[X53] AESG ←	Aesg 1*[X53] ←	Select channel group [X53] for input 1 to be decoded and embedded.
View channel group	[Esc] 1AESG ←	[X53] ← <i>In verbose modes 2 and 3:</i> Aesg 1*[X53] ←	View decoded channel group [X53].
Swap Audio Channels			
Set swap mode	[Esc] [X52]*[X53]*[X55] AESS ←	Aess [X52]*[X53]*[X55] ←	Set audio channel swap mode
View swap mode	[Esc] [X52]*[X53] AESS ←	[X55] ← <i>In verbose modes 2 and 3:</i> Aess [X52]*[X53]*[X55] ←	View audio channel swap mode.
KEY: [X50] = Output volume [X51] = Audio output format [X52] = SDI audio channel pair [X53] = SDI audio channel group [X55] = Swap channel [X58] = Audio input format 0 dB to -100 dB in 1.0 dB steps. 0 dB = default 1 = Dual mono, 2 = Stereo AES channel pair, 1 or 2. 1 = Group 1, 2 = Group 2 0 = Off, 1 = On 0 = None (input audio muted) 1 = Analog (on 5-pole captive screw connector) 2 = LPCM-2Ch (default) 3 = Multi-Ch			

Command	ASCII Command (Host to Scaler)	Response (Scaler to Host)	Additional Description
Advanced Configuration			
Test Pattern (persists through a power cycle)			
Set test pattern	[Esc] 1*[X22] TEST←	Test 1*[X22]←	Select test pattern [X22].
View test pattern	[Esc] 1TEST←	[X22]← <i>In verbose modes 2 and 3:</i> Test 1*[X22]←	View the current test pattern.
Freeze (Defaults to Off after a system reset or power cycle.)			
Enable freeze	1*1 F	Frz 1*1←	Freeze the input.
Disable freeze	1*0 F	Frz 1*0←	Unfreeze the input.
View	1F	[X10]← <i>In verbose modes 2 and 3:</i> Frz 1*[X10]←	Show the freeze status.
Input Aspect Ratio			
Set to Fill mode	[Esc] 1*1 ASPR←	Aspr 1*1←	Set the input to always fill the entire output raster (default).
Set to Follow mode	[Esc] 1*2 ASPR←	Aspr 1*2←	Set the input to preserve its native aspect ratio.
View aspect setting	[Esc] 1 ASPR←	[X39]← <i>In verbose modes 2 and 3:</i> Aspr 1*[X39]←	View input aspect ratio [X39].
Front Panel Lock (Executive Mode)			
Enable lock mode	1X	Exe1←	Lock the entire front panel (mode 1).
Disable lock mode	0X	Exe0←	Unlock front panel controls. All front panel adjustments can be made.
View lock mode status	X	[X29]← <i>In verbose modes 2 and 3:</i> Exe [X29]←	View current lock mode status [X29].
Video Signal Presence			
View video signal presence	[Esc] 0LS←	[X61]← <i>In verbose modes 2 and 3:</i> In00 • [X61]←	View input signal status [X61].
KEY: [X10] = Freeze or unfreeze the input [X22] = Test pattern [X29] = Front panel lock (executive mode) [X39] = Aspect ratio [X61] = Video signal status			
0 = Unfreeze, 1 = Freeze 0 = Off (default), 1 = Crop, 2 = Alternating pixels, 3 = Crosshatch, 4 = Color bars, 5 = Grayscale, 6 = Audio test (crop pattern, OSD text AUDIO TEST, outputs pink noise at 48 kHz, 24 bits, on channels 1 and 2) 0 = Unlock front panel (default), 1 = Lock front panel 1 = Fill : The input rate automatically fills the entire output raster (default). 2 = Follow : The input rate is displayed with its native aspect ratio. 0 = Video signal not detected, 1 = Video signal detected			
Resets			
Reset all device settings to factory default settings ²⁴	[Esc] ZXXX←	Zpx←	
Absolute system reset	[Esc] ZQQQ←	Zpq←	
NOTE: After an absolute system reset ([Esc] ZQQQ), the password reverts to extron.			
Absolute system reset, but retain all IP settings	[Esc] ZY←	Zpy←	
NOTE: This reset is the same as [Esc] ZQQQ except that it excludes IP settings such as IP address, subnet mask, and so on, in order to preserve communication with the device. This reset is recommended for after a firmware update.			

Command	ASCII Command (Host to Scaler)	Response (Scaler to Host)	Additional Description
Verbose Mode			
Set verbose mode	[Esc] [X31] CV ←	[X31] ←	Set the verbose mode for SIS command responses.
View verbose mode	[Esc] CV ←	[X31] ← <i>In verbose modes 2 and 3:</i> Vrb [X31] ←	View the current verbose mode.
Unit Name (TCP/IP Hostname)			
Set unit name	[Esc] [X32] CN ←	Ipn • [X32] ←	Specify a name for the unit.
Set unit name to factory default	[Esc] • CN ←	Ipn • [X33] ←	Reset the unit name to the factory default name.
View unit name	[Esc] CN ←	[X32] ← <i>In verbose modes 2 and 3:</i> Ipn • [X32] ←	View the name of the unit.
KEY: [X31] = Verbose mode [X32] = Unit name [X33] = Factory default name [X88] = Internal temperature in degrees Celsius [X89] = Over-temperature delta			
0 = None, 1 = Verbose mode (default), 2 = Tagged response for queries, 3 = Verbose mode and tagged response for queries A text string of up to 63 characters (A-Z, 0-9, and the hyphen (-) are permitted). • The first character must be a letter. • The last character cannot be a hyphen. • No blank or space characters are permitted. • No distinction is made between uppercase and lowercase letters. DSC 12G-HD A or DSC HD-12G A Response is two digits, padded with a zero. Number of degrees Celsius that the internal temperature is above or below the product-specific threshold, to one decimal place (preceded by a + or - sign)			
Passwords			
NOTE: The initial passwords set at the factory are the serial number of the unit. However, if the unit is reset (via the ZQQQ SIS command or the rear panel Reset button), this password reverts to extron.			
Set administrator password	[Esc] [X91] CA ←	Ipa • [X91] ←	Set the administrator password to [X91] .
View administrator password	[Esc] CA ←	**** ← <i>In verbose modes 2 and 3:</i> Ipa • **** ← or Ipa ←	View the administrator password.
Reset administrator password	[Esc] • CA ←	Ipa • **** ←	Reset the administrator password to the default (extron).
Set user password	[Esc] [X91] CU ←	Ipu • [X91] ←	Set the user password.
View user password	[Esc] CU ←	**** ← <i>In verbose modes 2 and 3:</i> Ipu • **** ← or Ipu ←	View the user password. If there is a valid password, the response is **** ←. If there is no password, the response is ←.
Reset user password	[Esc] • CU ←	Ipu • **** ←	Reset the user password.
KEY: [X91] = Password • Length is 1-128 characters. • All human-readable characters are permitted except . • The password cannot be a single space. • Passwords are case-sensitive. • A user password cannot be assigned if no administrative password exists. An E14 error code is returned.			
NOTE: If there is a valid password, the response is **** ←. If there is no password, the response is ←.			

Product Configuration Software

The Extron Product Configuration Software (PCS) is a Windows-based program for the DSC that provides a convenient way to configure the input and output, audio, and image settings. It also lets you save and recall presets and perform nearly all the other functions that can be accomplished via the front panel controls or SIS commands.

To download PCS from the Extron website, locate it on the Download Center page or go to the PCS product page.

Software Installation

To download PCS from the Extron website, locate it on the Download Center page or go to the PCS product page.

Software Download Center page

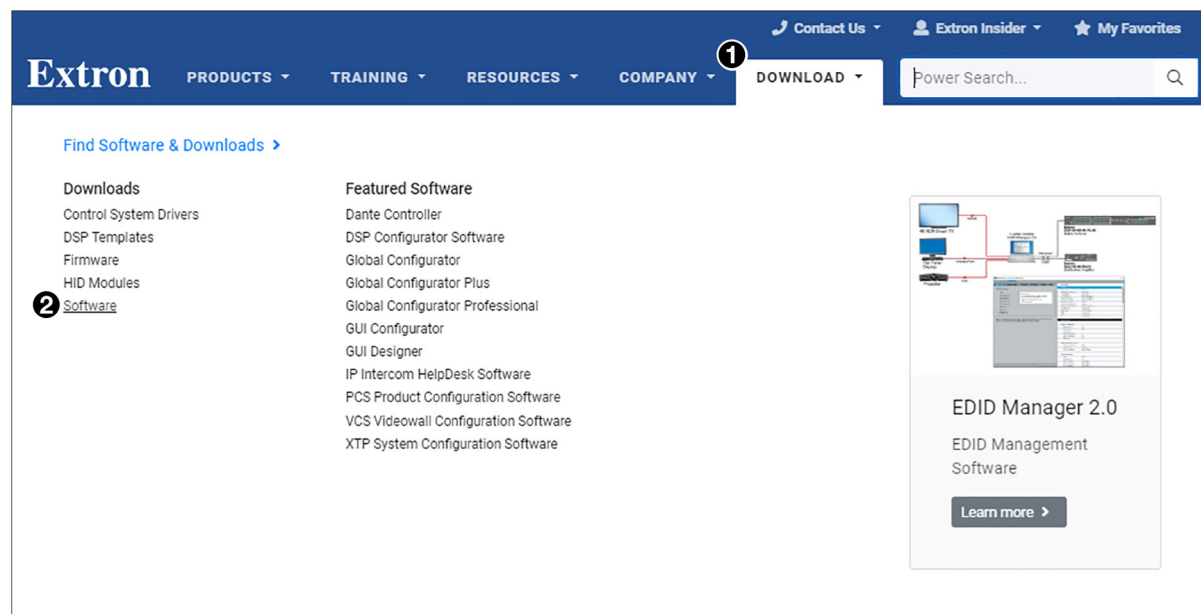


Figure 26. Download Center Page on the Extron Website

1. On the Extron website, select the **Download** tab (see figure 26, ①).
2. Move the pointer to the **Software** link (②) in the Downloads column and click it.

- On the Download Center screen, click on the down arrow in the **Search Software** box to see a list of the downloadable software (see figure 27, ❶). If you need an Extron Insider ID number to access this box, contact your Extron representative.

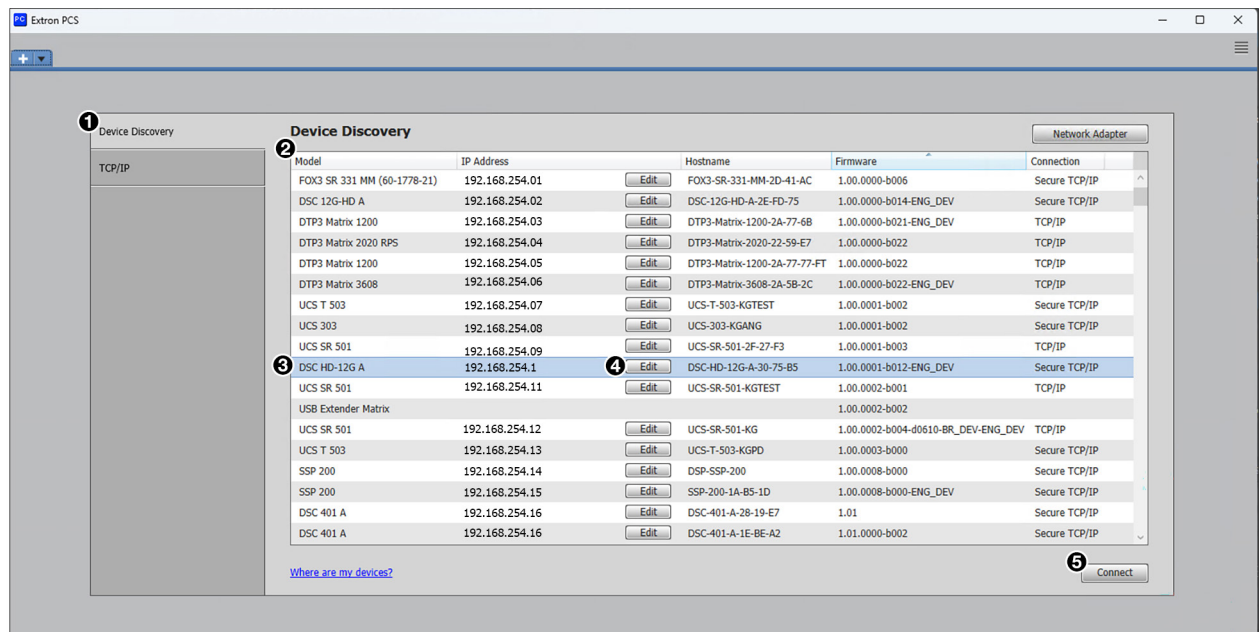


Figure 27. PCS Links on Software Download Center Screen

- From the drop-down list, select Product Configuration Software - PCS (❷).
- On the Product Configuration Software - PCS Download Center screen, click the link to the active PCS version (see figure , ❶). After a few seconds, a link to the VCS installer on your PC appears.

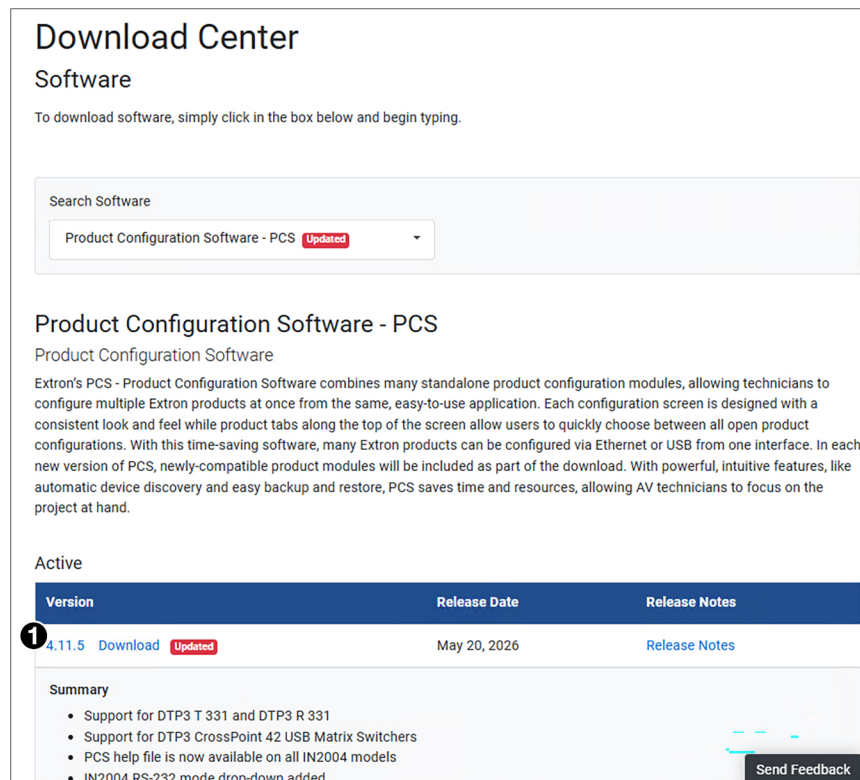


Figure 28. PCS Download Center Screen

6. Click the link to locate the installer file, then follow the instructions on the screen to complete the PCS installation.
7. On the login page that appears next, fill in the required information to log in to www.extron.com (if you need an Extron Insider ID number, see your Extron representative).
8. Follow the instructions on the subsequent screens to complete the software program installation. By default, the configuration program files are stored on your computer at:
C:\Program Files (x86)\Extron\Extron PCS.

If there is not already an Extron folder in your Program Files (x86) folder, the installer program creates it.

Software Connection

Open the PCS program from the Start menu or desktop shortcut. The Extron Product Configuration Software window opens with the Device Discovery panel open. Connect to the switcher using the Device Discovery panel or the TCP/IP panel.

Starting the Configuration Program

To run the PCS control program, do either of the following:

- Double-click on the EAF.exe file, located on the computer at
c:\Program Files [or Program Files(x86)]\Extron\Extron PCS.
- Access the program from the Start menu on the computer as follows:
 - a. Click the **Start** icon on the computer screen.
 - b. Click the **All** button on the Start screen.
 - c. From the Start menu, select **Extron/Extron-PCS**. The Extron Product Configuration Software window opens.

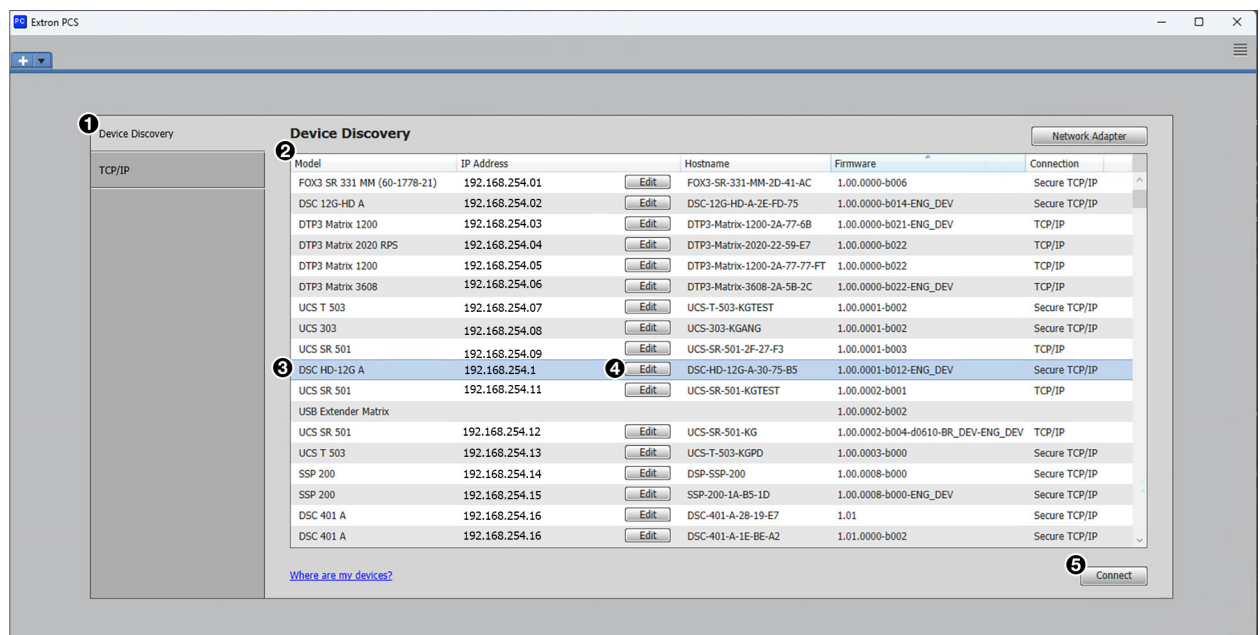


Figure 29. PCS Main Window

To sort the list of available devices:

1. Click the **Device Discovery** tab (see figure 29, ①).
2. Click the desired column heading (②) to sort it in ascending or descending order.

To connect to a device:

1. Click the **Device Discovery** tab (see [figure 29](#), **1** on page 59).
2. Select the desired device (**3**).
3. Click the **Connect** button (**5**). A new device configuration tab opens.

To edit communication settings from the Device Discovery panel:

1. Click the **Device Discovery** tab (**1**).
2. Click the **Edit** button of the desired device (**4**). The Communication Settings dialog box opens.
3. Finalize the settings in one of the following ways:
 - Click the **Apply** button to accept the changes and return to the Device Discovery panel.
 - Click the **Apply and Connect** button to accept the changes and connect to the selected device. A new device configuration tab opens.
 - Click the **Cancel** button to cancel any pending changes and return to the Device Discovery panel.

TCP/IP Panel

The TCP/IP panel connects PCS to a specific device through Ethernet. To access this panel, click the **TCP/IP** tab to the left of the Device Discovery panel (see [figure 30](#), **1**).

Figure 30. TCP/IP Panel

To connect directly to a connected device with a known IP address:

1. In the **IP Address/Hostname** field of the Target Device panel (**2**), enter the IP address of the desired device.
2. If required, enter the device password in the **Password** field.

NOTES:

- The factory configured passwords for all accounts on this device have been set to the device serial number. Passwords are case sensitive.
- If desired, select the **Show Characters** checkbox (below the **Port** field) to display the password characters.

3. In the **Port** field (see [figure 30](#), ③ on page 60), enter the port number of the desired device (the default is 4523).
4. Click the **Connect** button (⑥). A new device tab opens.

To connect indirectly to a device through a control processor via an AV LAN:

If the DSC scaler is connected to an AV LAN, it can be connected through a control processor such as an Extron IPCP, as follows:

1. In the **IP Address/Hostname** field of the Target Device panel (②) enter the IP address of the DSC unit on the AV LAN, as well as the port number and the password (if needed).
2. Select the **Enable Indirect Connection** checkbox (④).
3. In the **IP Address/Hostname** field of the IP Link Control Processor with AV LAN panel (⑤), enter the IP address of the control processor connected to the AV LAN.
4. Enter the password if required.
5. Click **Connect** (⑥). The TCP/IP window closes and the device configuration window opens.

Internal Web Pages

The DSC 12G-HD A/HD-12G A scalers feature an internal web server, displayed as a web page. This page enables monitoring and adjusting of certain settings on the DSC via a LAN or WAN connection. Use a web browser to view the pages on a PC connected to the scaler LAN port.

This section gives an overview of the internal web page, which is always available and cannot be erased or overwritten. Topics in this section include:

- [Accessing the Web Pages](#)
- [Web Page Components](#)

Accessing the Web Pages

1. Connect the scaler to a network using the rear panel RJ-45 LAN connector (see [figure 3 or 4](#), [G](#) on page 6), or the front panel USB-C CONFIG port (see [figure 9 or 10](#), [B](#) on page 10).

2. Open a web browser on a connected PC.

3. Enter the IP address of the device in the browser Address field.

NOTE: The default IP address is 192.168.254.254 (LAN port) or 203.0.113.22 (USB CONFIG port).

4. Press <Enter> on the computer keyboard. The product sign-in page opens (see figure 31).

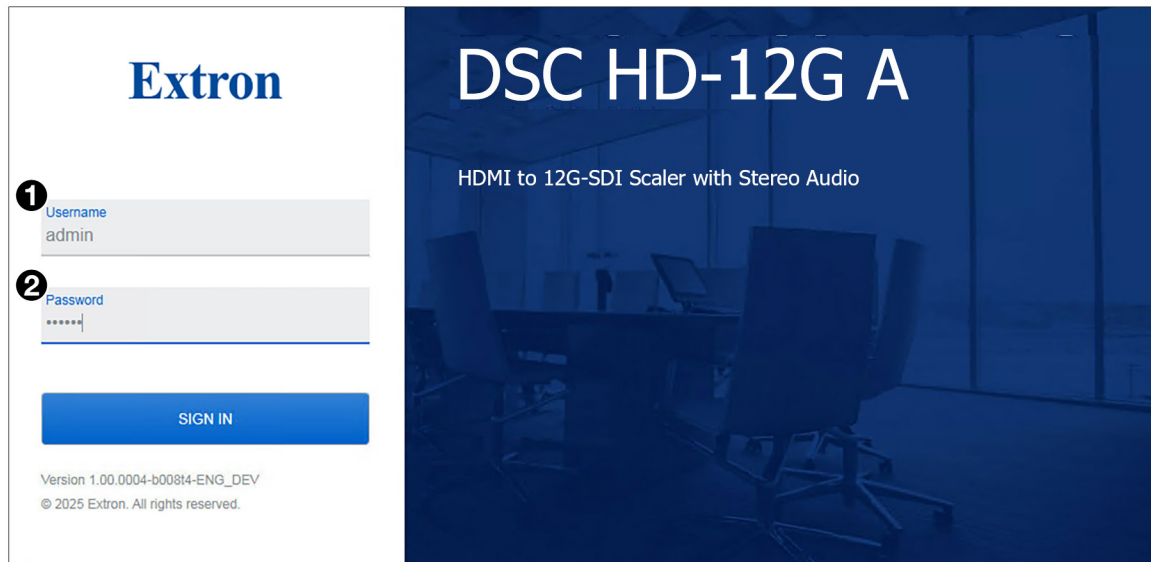


Figure 31. DSC HD-12G A Sign-in Page Example

5. Enter admin or user in the **Username** field ([1](#)).
6. If required, enter the password in the **Password** field ([2](#)):
 - If no password has been set since the unit was delivered, enter the unit serial number. This is usually located on a sticker on the side panel of the unit.

NOTE: The factory configured passwords for all accounts on this device have been set to the device serial number. The password defaults to extron after an absolute system reset. Passwords are case sensitive.

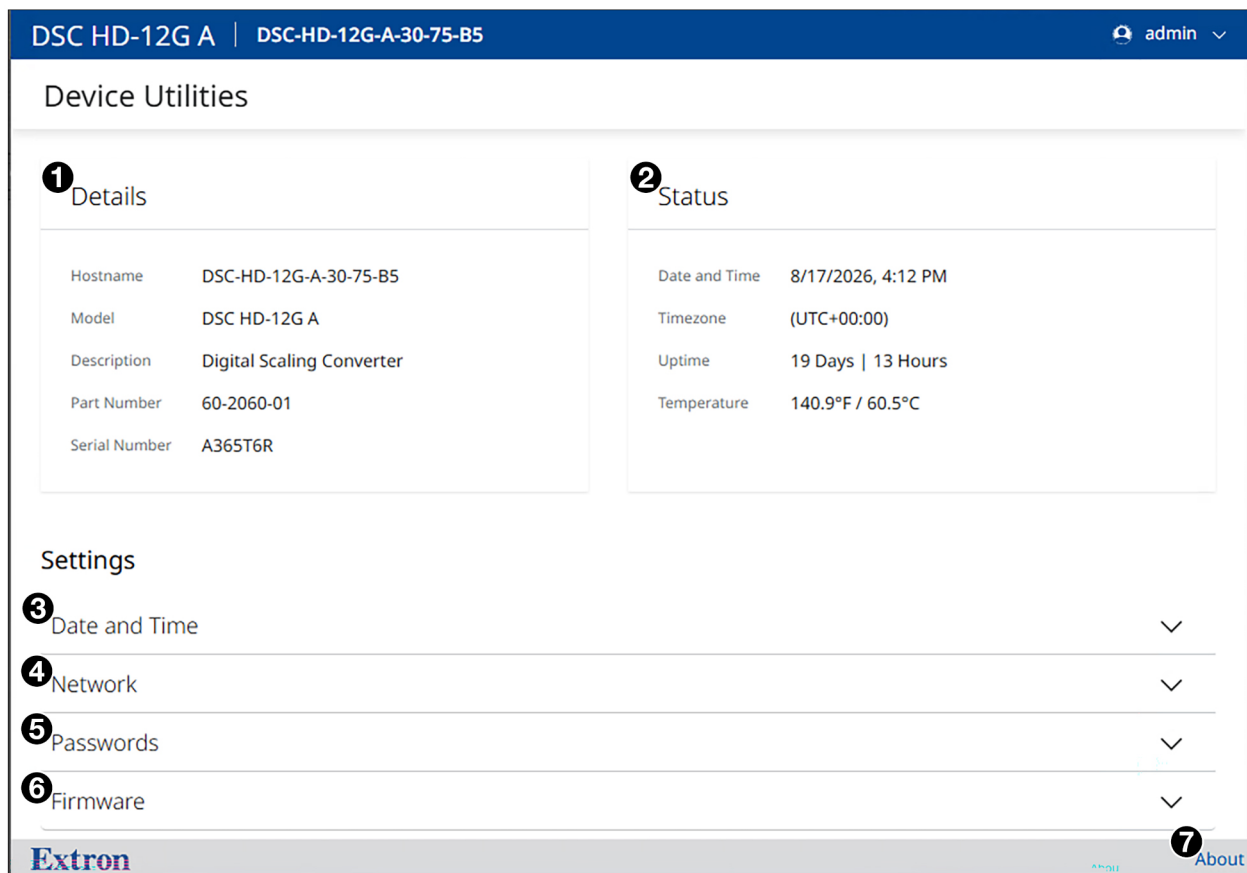
- If the unit has had an absolute system reset (the ZQQQ SIS command has been entered, see [Absolute system reset](#) on page 54), or if the factory password has been removed and the factory default password of extron will be set, click **Sign In** to display the internal web page.
- If a new password has been set, enter it and click **Sign In**.
- To change the password, see [Passwords Panel](#) on page 66.

Web Page Components

The DSC internal web page opens to the Device Utilities page (see figure 32). This page provides an overall, read-only description of the scaler and a view of its status. It also contains editable fields for setting the date and time, the network settings, the serial port settings, passwords, and firmware updates.

The panels in the accordion section that can be edited have an **EDIT** link to click to access the panel. To access each panel in the accordion, click the name or the arrow for each panel. To view general information about the DSC series, click the **ABOUT** link (7).

The internal web page does not automatically update. To see an updated page, click the **Refresh** button on the web browser.



- | | |
|-----------------------|-------------------|
| 1 Details Panel | 5 Passwords Panel |
| 2 Status Panel | 6 Firmware Panel |
| 3 Date and Time Panel | 7 About Link |
| 4 Network Panel | |

Figure 32. DSC HD-12G A Internal Web Page

Details Panel

The Details panel (see [figure 32](#), [1](#) on page 63) displays the device name, a brief product description, the part number, and the serial number.

Status Panel

The Status panel ([2](#)) displays the current date, time, time zone, internal temperature in degrees Fahrenheit and Celsius, and the amount of time the device has been running (Uptime).

Date and Time Panel

The Date and Time panel ([3](#)) displays the mode used to set the date and time, current date, and time. There are three options to set the date and time: manual mode, sync to NTP (Network Time Protocol) servers, and sync to the connected PC.

To set the date and time in Manual mode:

1. Click **EDIT** in the Date and Time panel. The editable fields are displayed (see [figure 34](#)).
2. Select the **Manual** radio button.
3. Edit the **Date/Time** as desired by typing in the date and time or using the calendar and clock. Click the calendar or clock icon in the **Date | Time** field to display the windows shown below.

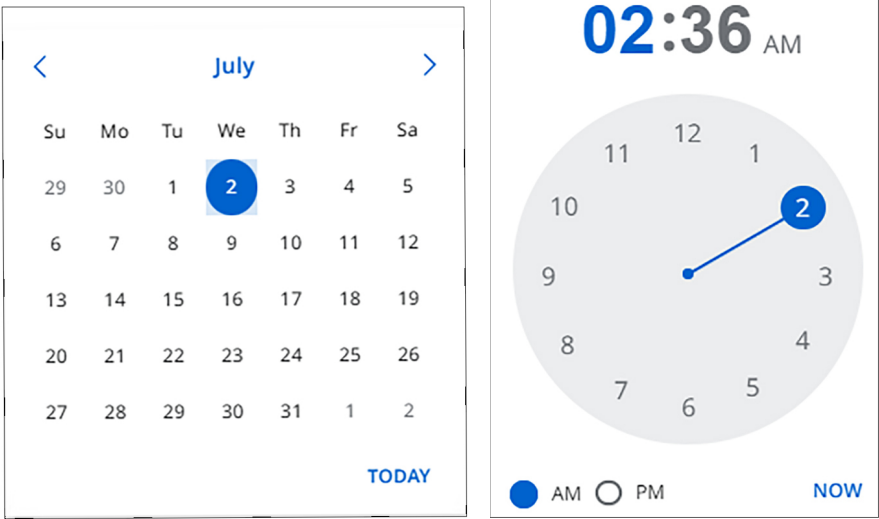


Figure 33. Datepicker and Timepicker Windows

4. Edit the **Time Zone** as desired, using the drop-down list.
5. When finished editing, click **SAVE** to confirm your changes, or **CANCEL** to close the window without making changes.

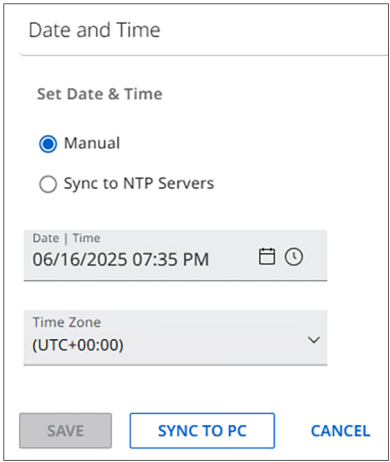


Figure 34. Manual Mode

To set the date and time in Sync to NTP Servers mode:

1. Click **EDIT** in the Date and Time panel. The editable fields are displayed.
2. Select the **Sync to NTP Servers** radio button (see figure 35).
3. Select the desired time zone from the **Time Zone** drop-down list.
4. Click in an **NTP Server** field and enter the IP address or host name of a desired NTP Server.
5. When finished editing, click **SAVE** to confirm your changes or **CANCEL** to close the window without making changes.

To sync the date and time with the PC:

Click **SYNC TO PC** to set the date and time according to the PC.

The screenshot shows the 'Set Date & Time' window. At the top, there are two radio buttons: 'Manual' (unselected) and 'Sync to NTP Servers' (selected). Below this is a 'Time Zone' dropdown menu showing '(UTC+00:00)'. There are three input fields for NTP servers: 'NTP Server #1' with the value 'time.nist.gov', 'NTP Server #2' (empty), and 'NTP Server #3' (empty). Below these is a 'Current Date & Time' field showing 'Jun 16, 2025, 7:35:58 PM'. At the bottom are 'SAVE' and 'CANCEL' buttons.

Figure 35. Sync to NTP Servers Mode

Network Panel

In the Network panel (see [figure 32](#), 4 on page 63), view the MAC address, edit the device name, set the IP address, subnet mask, and DNS server for your DSC unit, and turn DHCP **On** and **Off**.

To set the IP addresses:

1. Click **EDIT** in the Network panel. The editable fields are displayed (see figure 36).
2. Enable or disable DHCP. When DHCP is enabled (On), the unit configures its IP address and other network settings from the DHCP server. The default is Off.

To enable DHCP, select **DHCP** from the Configure IP drop-down list.

NOTE: When DHCP is enabled, none of the other network fields can be edited.

3. Edit network fields as desired. To set network fields — **IP Address**, **Subnet Mask**, **Gateway** address, and **DNS Server** addresses — do the following:
 - a. From the Configure IP drop-down list, select **Static**.
 - b. Click in the desired fields and enter the addresses.
4. When finished editing, click out of the field and **SAVE** to confirm your changes or **CANCEL** to close the window without making changes.

The screenshot shows the 'Network' configuration window. It has a title bar 'Network' with a close button. The 'Device Name / Hostname' field contains 'DSC-12G-HD-A-2E-FD-75'. Below it are 'SAVE' and 'CANCEL' buttons. The 'MAC Address' field contains '00-05-A6-2E-FD-75'. The 'Configure IP' dropdown is set to 'Static'. Below this are fields for 'IP Address' (192.168.254.254), 'Subnet Mask' (255.255.255.0), and 'Gateway' (0.0.0.0). The 'DNS Server' section has three fields: 'DNS Server 1' (10.1.0.3), 'DNS Server 2' (10.1.0.7), and 'DNS Server 3' (empty). There is a 'Search Domain' field with 'extron.com'. At the bottom are 'SAVE' and 'CANCEL' buttons.

Figure 36. Network Panel

Passwords Panel

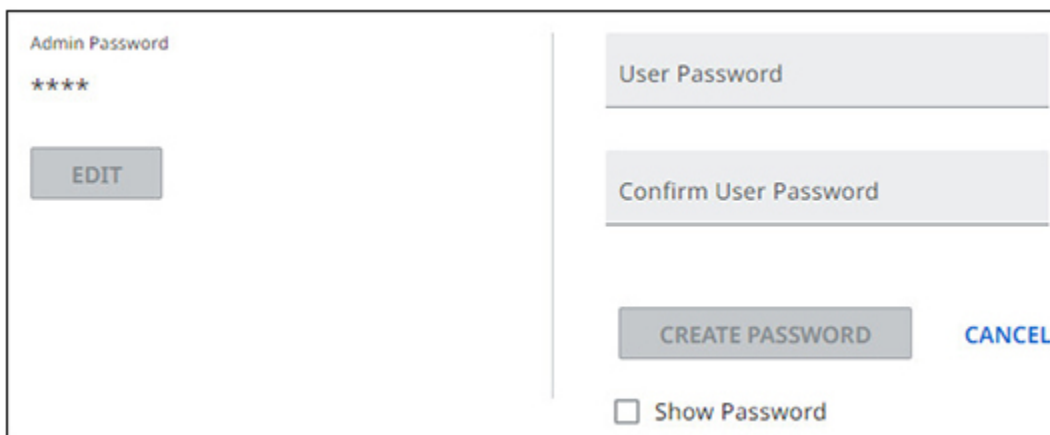
The Passwords panel (see [figure 32](#), ⑤ on page 63) shows whether Admin and User passwords have been set and provides options to set them. It does not display the actual password.

NOTE: The following rules apply to passwords:

- Length is 1-128 characters.
- All human-readable characters are permitted except | .
- The password cannot be a single space.
- Passwords are case-sensitive.
- The factory configured passwords for all accounts on this device have been set to the device serial number. In the event of a complete system reset, the passwords convert to the default, which is **extron** for the Admin account.

To assign a administrator and user passwords:

1. In the Passwords panel, click **EDIT**. The editable fields are displayed (see figure 37).



The screenshot shows the Passwords Panel interface. On the left, under the heading "Admin Password", there is a field containing five asterisks (*****) and an "EDIT" button below it. On the right, under the heading "User Password", there is a field for the password and a "Confirm User Password" field below it. At the bottom right, there is a "CREATE PASSWORD" button and a "CANCEL" link. At the bottom left, there is a checkbox labeled "Show Password".

Figure 37. Passwords Panel

2. Click in the **Admin Password** field and enter the administrator password.
3. Click in the **Confirm Admin Password** field and enter the same password from the Admin Password field.
4. To assign a user password, repeat steps 2 and 3 in the User panel.

NOTE: The User Password fields become editable when a Admin Password is set.

5. When finished, click **SAVE** to set the passwords. To close the window without saving a password, click **CANCEL** or the X in the upper-right corner.

NOTE: Passwords can be changed but they cannot be removed entirely. The DSC must have passwords at all times. These fields cannot be blank.

Firmware Panel

The Firmware panel (see [figure 32](#), ⑥ on page 63) displays the current firmware version and the date it was last updated. Update the firmware on the DSC series scaler from this panel (see [Downloading Updated Firmware](#) on page 71 to download the firmware files).

To update firmware:

1. In the Firmware panel, click the **Select File** button (see [figure 38](#)).
2. In the Open dialog box that appears, browse to locate the new firmware file on your computer (by default the file is stored at C:\Program Files (x86)\Extron\Firmware\DSC 12G-HD and HD-12G after being downloaded from the [Extron web page](#)).

NOTE: Firmware files for the DSC series have a .eff extension. Do not attempt to load any other file types.

3. Double-click the firmware file name. The Open window closes, and the selected firmware file name appears in the Update Firmware panel on the web page (see [figure 39](#), ①).
4. Click **INSTALL** (②) to begin.
5. To cancel the update, click the X button in the Update Firmware panel.

During the updating process, a window appears in the middle of the screen, showing the progress of the update: Initializing, Installing the Firmware, and Rebooting Device.

When the update is done, the message window closes and the message Firmware Upload Complete displays near the top of the screen. The new firmware filename displays beside Version in the Firmware panel.

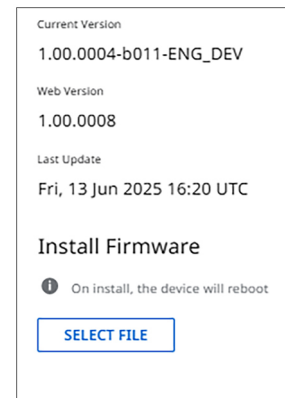


Figure 38. Firmware Panel

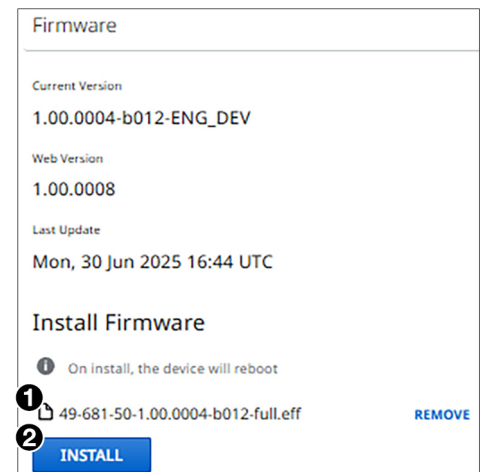


Figure 39. Firmware Install Panel

About Link

The **About** link, ⑦, located in the bottom right corner of the web page, opens a screen containing general information about the DSC and links to the product licenses and patents. To expand a panel on the About screen, click the **down arrow** button at the right of the panel (see figure 40).

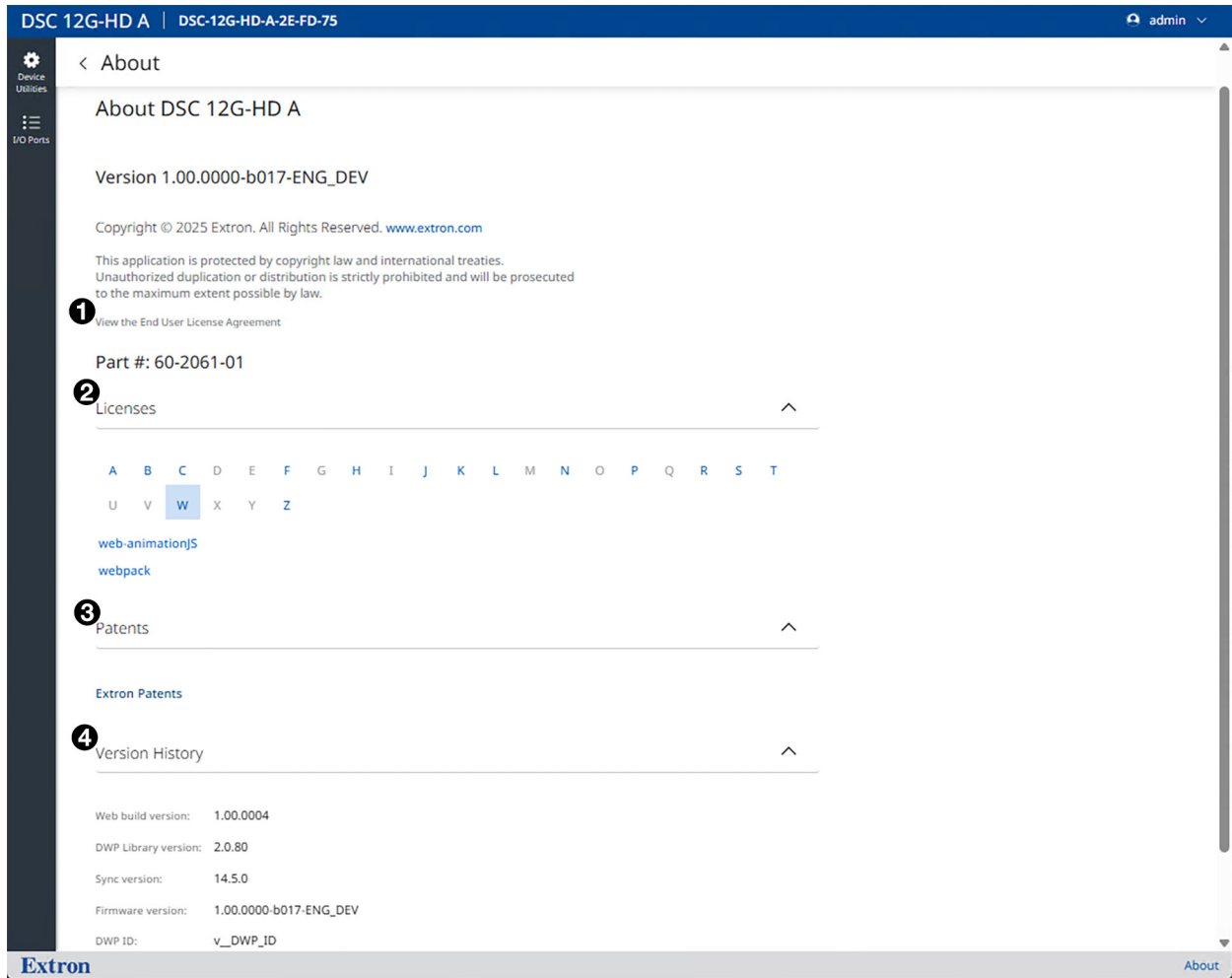


Figure 40. About Page, Expanded Panels

- ① **View the End User License Agreement link** — Opens the EULA for online viewing.
- ② **Licenses panel** — Expand the panel to display a set of alphabet links. To view the desired license, click its initial letter, then the license name.
- ③ **Patents panel** — Expand the panel to display a link to the Extron patents.
- ④ **Version History panel** — Expand the panel to view the firmware version, web build version, and other version information.

Reference Information

This section contains information on mounting the DSC scaler. See <https://www.extron.com> for available Extron mounting kits and ordering information. The DSC 12G-HD A and DSC HD-12G A scalers can be mounted by any of the methods described below. To mount the DSC using an Extron mounting kit, see the instructions provided with the kit.

This section provides reference or supplemental information. Topics in this section include:

- [Rack Mounting](#)
- [Furniture Mounting](#)
- [Best Practices for Cleaning Your Extron Products](#)
- [Downloading Updated Firmware](#)

Rack Mounting

Attach the DSC to a standard 19-inch rack shelf.

UL Guidelines for Rack Mounting

The following Underwriters Laboratories (UL) guidelines pertain to the safe installation of the DSC 12G-HD A or DSC HD-12G A in a rack.

CAUTION:

- **Elevated operating ambient temperature** — If the equipment is installed in a closed or multi-unit rack assembly, the operating ambient temperature of the rack environment may be greater than room ambient temperature. Therefore, install the equipment in an environment compatible with the maximum ambient temperature (TMA = +122 °F, +50 °C) specified by Extron.
- **Reduced air flow** — Install the equipment in a rack so that the amount of air flow required for safe operation of the equipment is not compromised.
- **Mechanical loading** — When mounting the equipment in the rack, ensure that uneven mechanical loading does not cause a hazardous condition.
- **Circuit overloading** — When connecting the equipment to the supply circuit, consider the effect that circuit overloading might have on overcurrent protection and supply wiring. Consider equipment nameplate ratings when addressing this concern.
- **Reliable earthing (grounding)** — Maintain reliable grounding of rack-mounted equipment. Pay particular attention to supply connections other than direct connections to the branch circuit (for example, use of power strips).

Consignes UL Pour le Montage en Rack

Les consignes UL (« Underwriters Laboratories ») suivantes concernent l'installation en rack d'un boîtier DSC 12G-HD A ou DSC HD-12G A :

ATTENTION :

- **Température ambiante élevée** — En cas d'installation de l'équipement dans un rack fermé ou composé de plusieurs unités, la température du rack peut être supérieure à la température ambiante. Par conséquent, il est préférable d'installer l'équipement dans un environnement qui respecte la température ambiante maximale (Tma) spécifiée par Extron.
- **Réduction du flux d'air** — Si l'équipement est installé dans un rack, veillez à ce que le flux d'air nécessaire pour un fonctionnement sécurisé de l'équipement soit respecté.
- **Charge mécanique** — Installez l'équipement en rack de manière à éviter toute situation dangereuse causée par le déséquilibre de la charge mécanique.
- **Surcharge électrique** — Lorsque vous connectez l'équipement au circuit d'alimentation, observez la connexion de l'équipement et étudiez les effets possibles d'une surcharge du circuit sur les protections contre les surintensités et les conducteurs d'alimentation. Consultez à cet égard les indications de la plaque d'identification de l'équipement.
- **Mise à la terre** — Assurez-vous que l'équipement est correctement mis à la terre. Accordez une attention particulière aux connexions électriques autres que les connexions directes au circuit de dérivation (ex. : les multiprises).

Furniture Mounting

- **Under-furniture mounting** — Mount the DSC under the surface of a desk, table, or podium.
- **Through-furniture mounting** — Mount the DSC through a table or desk in which a hole has been cut to enable access to the front panel.
- Free-standing

Attach the four rubber feet to the bottom of the DSC at the four corners and place the unit where desired.

Best Practices for Cleaning Your Extron Products

Depending on the device, application, and location, there may be times when it becomes necessary to clean your Extron product. Frequently touched devices, such as touchscreens and button panels, require regular cleaning to ensure their surfaces remain sanitary. Plastic surfaces and cosmetic finishes can be damaged by long term exposure to chemicals. Therefore, Extron recommends the following guidelines when cleaning our products.



All Extron products can be safely cleaned with:

- 70% concentration or higher Isopropyl Alcohol
- Disinfectant cleaners that:
 - Are non-ammonium based (for example, contains no ammonium chloride).
 - Contain 2% or less sodium hypochlorite (for example, 2% bleach, 98% water).

Regardless of the device, it is important to follow these general guidelines when cleaning:

1. If possible, unplug the device.
2. Spray the cleaner on a lint-free cloth until the cloth is damp.
3. Do not spray the cleaner directly onto the product.
4. Gently clean the product surface using the cloth.

Your health and safety are an Extron priority. Keeping devices clean, especially those in high-traffic environments and high-use applications, is a crucial step in minimizing the spread of infections. Please contact Extron if you have any questions about the guidelines outlined in this section or if you have a question about cleaning a specific Extron product.

Downloading Updated Firmware

Should a new version of the firmware become available, download it from the Extron website as follows:

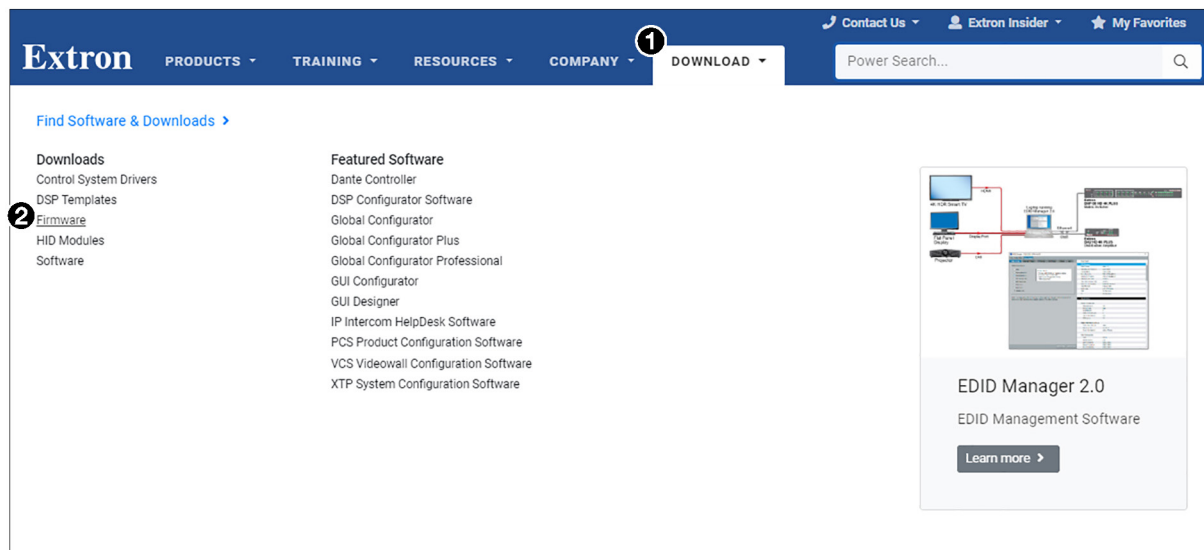


Figure 41. Downloading Firmware from the Extron Website

1. On the [Extron web page](#), click the **Download** tab (see figure 41, ①).
2. Move the pointer to the **Firmware** link (②) in the Downloads column and click it.
3. On the Software Download Center screen, click on the down arrow in the Search Firmware box (see figure 42, ①), and select **DSC 12G-HD** or **DSC HD-12G** from the drop-down list. If you need an Extron Insider ID number to access this box, contact your Extron representative.

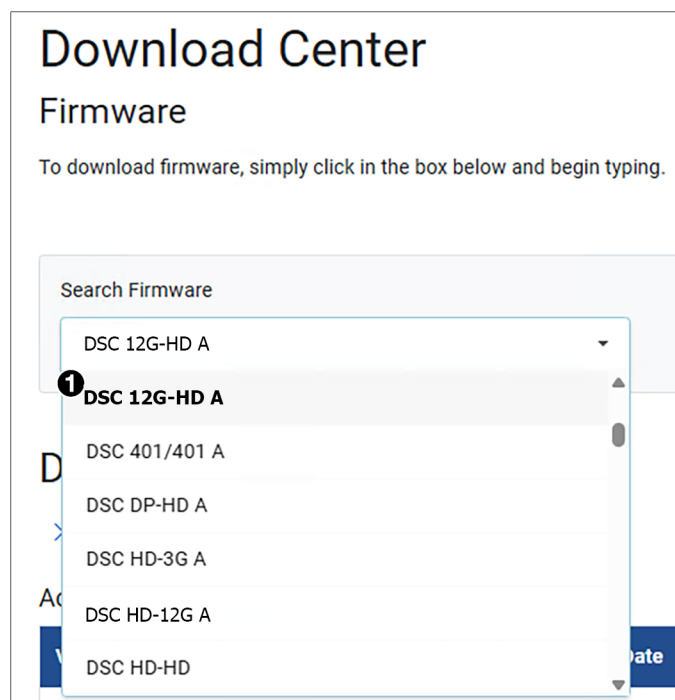


Figure 42. Search Firmware Drop-down List

4. Ensure the available firmware version is a later version than the current one on the device.

NOTE: The firmware release notes provide details about the changes between different firmware versions. The file can be downloaded from the same page as the firmware.

5. Click the link to the active firmware version (see figure 43, ①).

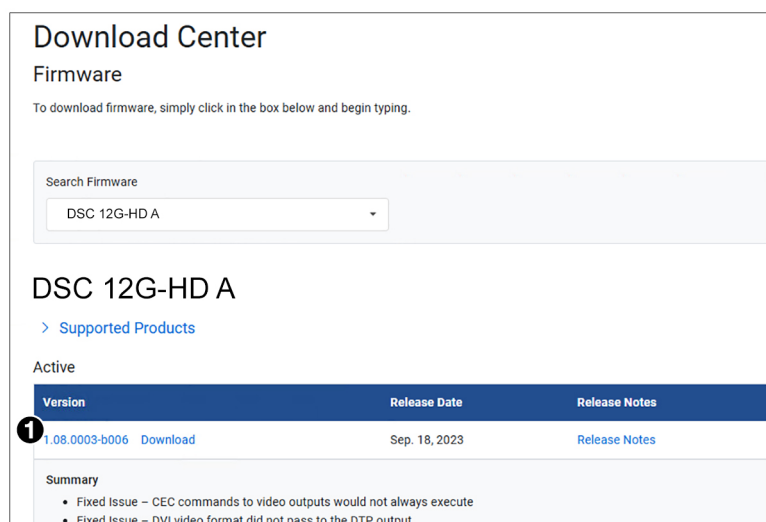


Figure 43. Download Link

After a few seconds, a link to the firmware installer file in your Downloads folder appears.

6. Click the link to the installer file, then follow the instructions on the screens to install the firmware.

By default, the firmware file is stored on the computer in the Downloads folder at \Desktop\This PC. From there, if desired, move it to another storage location on the C: drive.

Extron Warranty



Powered Product Warranty

Extron warrants its powered products against any defects in materials and workmanship for a period of three years from the date of invoice. In the event of a malfunction during the warranty period, Extron will repair or replace a product to its original operating condition.

To assure the highest level of service, a return authorization number must be obtained from Extron before products are returned for service. Products must be shipped to Extron, prepaid along with proof of purchase ONLY after obtaining a Return Authorization (RA) number from the Extron Customer Support department.

This statement of policy is in lieu of any other policy expressed or implied and no representative or person is authorized to assume any other liability or adopt any other policy for Extron Electronics without our written consent.

Powered Product Warranty Exceptions

Everlast™ Power Supplies — Extron warrants Everlast power supplies against any defects in materials and workmanship for a period of seven years from the date of invoice. In the event of a malfunction during the warranty period, Extron will repair or replace the power supply to its original operating condition. Extron engineers will examine the returned product and determine whether the Everlast Power Supply Warranty or Powered Product Warranty applies.

Speakers — Extron warrants Flat Field®, SoundField®, SpeedMount®, Column Array, and System *INTEGRATOR*® speakers against any defects in materials and workmanship for a period of five years from the date of invoice.

Touchscreens — Extron warrants touchscreen display and overlay components against any defects in materials and workmanship for a period of one year from the date of invoice.

Annotator 300 — Extron warrants the Annotator 300 against any defects in materials and workmanship for a period of five years from the date of invoice.

Non-Powered and Cable Products - Lifetime Limited Performance Warranty

Extron warrants that its non-powered products and cable products will be free from defects in material and workmanship for as long as you or your customer owns the product. All Extron non-powered products and cables are designed and engineered to meet and exceed performance specifications. If, at anytime, the product fails due to manufacturer defect, Extron will repair or replace the product to ensure that it meets original performance specifications. Reduced performance due to normal wear and tear, or damages caused by misuse or negligence will not be covered. Extron will test and evaluate all non-powered and cable products claimed defective.

To assure the highest level of service, a return authorization number must be obtained from Extron before products are returned for service. Products must be shipped to Extron, prepaid along with proof of purchase ONLY after obtaining a Return Authorization (RA) number from the Extron Customer Support department.

This statement of policy is in lieu of any other policy expressed or implied and no representative or person is authorized to assume any other liability or adopt any other policy for Extron Electronics without our written consent.

Non-Powered and Cable Product Warranty Exceptions

Cable Cubby, Hideaway Surface Access Enclosures and Retractors — Extron warrants Cable Cubby cable access enclosures, HSA Hideaway Surface Access enclosures, and Retractor cable retraction modules for a period of three years from the date of invoice.

Active Cables and Active Adapters — Extron warrants active cables and active adapter cables for a period of three years from the date of invoice.

Cable Termination Tools and Dies — Extron warrants cable termination tools for a period of three years from the date of invoice, excluding the die.

General

This **Limited Warranty** does not apply if the fault has been caused by misuse, improper handling care, electrical or mechanical abuse, abnormal operating conditions or non-Extron authorized modification to the product. Extron Electronics makes no further warranties either expressed or implied with respect to the product and its quality, performance, merchantability, or fitness for any particular use. In no event will Extron Electronics be liable for direct, indirect, or consequential damages resulting from any defect in this product even if Extron Electronics has been advised of such damage. Please note that laws vary from state to state and country to country, and that some provisions of this warranty may not apply to you.

Return Information

Units must be returned insured, with shipping charges prepaid. If not insured, you assume the risk of loss or damage during shipment. Returned units must include the serial number and a description of the problem, as well as the name of the person to contact in case there are any questions.

NOTE: To assure the highest level of service, a return authorization number must be obtained from Extron before products are returned for service. Products must be shipped to Extron, prepaid along with proof of purchase **only** after obtaining a Return Authorization (RA) number from the Extron Customer Support department.

Please contact Extron to receive an RA (Return Authorization) number:

USA: 714.491.1500 or 800.633.9876

Asia: 65.6383.4400

Europe: 31.33.453.4040 or 800.3987.6673

Japan: 81.3.3511.7655

Africa and Middle East: 971.4.299.1800

China: 4000.398766

Worldwide Headquarters: Extron USA West, 1025 E. Ball Road, Anaheim, CA 92805, 800.633.9876